

Separation and Purification Technology

Review and Development of an Easy-to-Use Machine Learning Model for Pollutant Gas Solubility in Green solvent of Ionic Liquids

--Manuscript Draft--

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Keywords:	Genetic programming; Global warming; ionic liquids; machine learning; Pollutant gases
Abstract:	The increasing release of greenhouse gases (GHGs) like CO₂, CH₄, N₂O, and industrial contaminants (indirect GHGs) such as SO₂ and H₂S has prompted significant global worries due to their role in climate change, air pollution, and harm to the environment. Ionic liquids (ILs) as green solvents have emerged as promising alternatives to traditional solvents because of their minimal volatility, high thermal stability, and adjustable physicochemical characteristics. Yet, limited gas solubility data in ILs is hindering their applications in carbon capture and air pollution control. Machine learning (ML) is a powerful tool for modeling and simulating the solubility of polluting gases in ILs. This research aims to critically review recent progress in ML modeling of pollutant gas removal by ILs. More importantly, a new ML model of genetic programming (GP) was developed to generate a simple and accurate mathematical equation to predict the solubility of SO₂, CH₄, N₂O, CO, H₂S and CO₂ in ILs, using a large dataset (3209) for different gas-IL systems. Using temperature, pressure, and structural related parameters of ILs and gases as input parameters, the model achieved a high accuracy ($R^2 > 0.97$). Finally, a simple Excel method for calculating gas solubility has been created for prediction and modeling purposes.

Response Letter:

Thank you so much for the reviewers' comments. We are grateful for the valuable suggestions and comments of the reviewers. We have addressed all comments in the revised manuscript; all replies are in red in the revised manuscript.

Reviewer #1

Manuscript No.: SEPPUR-D-25-12636

Title: Review and Development of an Easy-to-Use Machine Learning Model for Pollutant Gas Solubility in Green solvent of Ionic Liquids

Comments: This manuscript presents a work of two parts: first, a comprehensive literature review of machine learning (ML) models applied to the prediction of pollutant gas solubility in ionic liquids (ILs); and second, the development of a new, multi-gas predictive model using Genetic Programming (GP). The authors have compiled a substantial dataset comprising 3209 data points for six industrially relevant gases (CO₂, H₂S, SO₂, CH₄, N₂O, CO). Using this dataset, they have developed explicit mathematical equations to predict gas solubility as a function of temperature, pressure, and key physicochemical properties of the gases and ILs. The central claims of the work are that this new model is both highly accurate, achieving a coefficient of determination R² greater than 0.97, and "easy-to-use" through a "simple Excel method".

Some comments for improvements of the manuscript:

1. References 64 and 65 were published in 2019 and 2018, respectively, which are inconsistent with the description of "latest studies" in the original text. Please update them to references published within the most recent five years or revise the description of "latest studies" in the original text.

Author response: Thank you for your valuable comment. The authors updated the references and added some of the recent studies in this regard. Please see the highlighted section on Page 5:

While several of the latest studies have incorporated analytical methods [64-68].

2. In comparison with the authors' previous work (Amirkhani et al.'s CSA-LSSVM model), which was used to predict the solubility in the exact same six ionic liquids, that study employed a dataset of similar scale (3060 data points, compared to 3209 in the current work) with an overall Average Absolute Relative Deviation (AARD) of 5.61%. The performance of the current Gaussian Process (GP) model is presented separately by gas type, with AARD values ranging from 4.65% for CO to 17.07% for N₂O. Therefore, it remains unclear whether the new GP model offers any significant improvement in prediction accuracy over their previous LSSVM model; in the case of N₂O, its performance appears to be much worse. The authors are required to add a direct and quantitative comparison between the performance of the current GP model and the CSA-LSSVM model in the manuscript. This comparison should be based on the same evaluation metrics and conducted on a gas-by-gas basis.

Author response: Thank you for your valuable comment. Comparison with CSA-LSSVM have been presented. Please see the highlighted section in Page 37:

Table S7 demonstrates the comparison of the statistical parameters of this study and the previous study [132], which used a CSA-LSSVM model to predict gas solubility in SO₂, CH₄, N₂O, CO, H₂S and CO₂. The prediction accuracy of CSA-LSSVM is higher in all gases except SO₂, in which GP showed higher accuracy. However, the GP model is easier to use than complex AI models.

3. A key innovation of the GP method lies in its ability to generate explicit mathematical equations, which contrasts with "black-box" models such as ANN or LSSVM. However, the manuscript fails to fully leverage this characteristic. The Discussion section does not utilize these complex equations to derive any new physical insights or understanding regarding gas-ionic liquid systems.

Author response: Thank you for your valuable comment. The related discussion has been added to Section 3.3. Please see the highlighted section on Page 36.

Although GP generates explicit mathematical formulas, the resulting expressions tend to be complex due to the need to account for the highly nonlinear relationship between gas solubility and factors like temperature, pressure, and the critical properties of ionic liquids. Consequently, the symbolic relationships produced by GP do not translate easily into straightforward, interpretable physicochemical principles. Instead, their significance lies in offering clear, equation-based predictions while still demonstrating physically meaningful behaviour. This is illustrated in Figure 2, where the GP model accurately captures the essential thermodynamic patterns. As shown in Figure 2, the gas solubility in ILs was increased by increasing pressure and decreasing temperature. However, there is an exception for CO solubility in [BMIM][MeSO₄] [154], which was increased by increasing temperature, which has been mentioned as a rare phenomenon [154]. Thus, although the GP equations do not provide new mechanistic insights into the interactions between gas and ionic liquids, they effectively encapsulate and reflect the anticipated physical trends across a diverse range of systems with great precision.

4. The complexity of the generated equations may render them less user-friendly than a trained black-box model. A stronger and more persuasive argument must be constructed to justify why an explicit (yet extremely complex) equation is superior to a well-performing black-box model. Specifically, what practical advantages does this model offer that previous models cannot provide?

Author response: Thank you for your valuable comment. The related discussion has been responded to the section 3.2. Please see the highlighted section (second paragraph in section 3.2) on Page 30.

Solution quality, primarily by utilizing the MSE, which is targeted for minimization [152]. While the mathematical intricacies of the correlations generated by GP are complicated, they present several advantages that black-box models fail to deliver. To begin with, the explicit equation enables users to calculate gas solubility directly without the need for training datasets, specialized machine learning software, or model retraining, thus making the correlation easily portable for use in industrial applications or offline scenarios (such as spreadsheets, process simulators, and numerical codes). Additionally, the functional format offers full transparency, allowing users to examine how temperature, pressure, and IL characteristics influence the model, which is not achievable with ANN or LSSVM frameworks. Moreover, the GP equation can be integrated into design calculations or utilized within process optimization frameworks as a closed-form function, a capability that is largely not possible with most black-box methodologies. Consequently, despite its mathematical complexity, the GP correlation provides greater interpretability, reproducibility, and practical applicability compared to traditional black-box models.

5. The manuscript repeatedly refers to a "simple Excel method" as crucial for model usability, yet this method has never been provided, described, or explained. The authors should clearly describe how to use this Excel tool for reviewers' evaluation and for all readers' access after publication.

Author response: Thank you for your valuable comment. The Excel file has been uploaded as supplementary material. Please see the related information in the highlighted section on Page 31.

A simple-to-use Excel file was developed to calculate the solubility using the input parameters. The users can put the data in each cell and calculate the solubility easily.

6. The authors should add a discussion on model complexity, address the possibility of overfitting, and justify the need for such a complex functional form rather than dismissing it as merely a product of the training process.

Author response: Thank you for your valuable comment. The related discussion has been responded to the section 3.3. Please see the highlighted section (first highlighted paragraph) on Page 35.

While the final GP correlations may appear mathematically intricate, this intricacy is indicative of the significant nonlinearity inherent in gas–IL systems rather than a result of overfitting. The model was developed using a substantial and varied dataset, and the strong alignment of R^2 , RMSE, and %AARD between the training and testing subsets suggests that the produced expressions generalize effectively without excessive variance. GP enhances the model structure only when additional terms lead to a discernible decrease in prediction error, indicating that the complexity is not arbitrary but stems from the physical and thermodynamic behaviors captured in the data. Importantly, even though the GP equations are lengthy, they provide a closed-form correlation that can be readily utilized in spreadsheets, engineering simulators, or design tools without the need for ML software, stored weight matrices, or retraining. Therefore, the

complexity of the final expression is warranted by its precision, portability, and clarity, offering benefits that conventional black-box models do not provide.

7. The rationale for selecting the hyperparameters of the GP algorithm is insufficient, as it is merely described as "obtained by trial and error". This approach is subjective, non-reproducible, and cannot guarantee that the optimal model configuration has been found.

Author response: Thank you for your valuable comment. The related discussion has been responded to the section 3.2. Please see the highlighted section on Page 31.

To improve the GP model and remove arbitrary hyperparameter selections, a systematic multi-stage tuning strategy was utilized. First, wide parameter ranges were defined based on values from previous symbolic regression studies (population size 50–600, tree depth 6–12, genes 2–10, generations 1500–3000). The GP was then sequentially trained using various configurations within these specified ranges. Each configuration was evaluated based on (i) the consistency of R^2 , RMSE, and %AARD between training and testing, (ii) the stability of convergence across multiple runs with different random seeds, and (iii) the simplicity of the model to avoid unnecessary equation complexity. Hyperparameters were selected only when they provided stable convergence and similar accuracy across repeated trials, thus guaranteeing reproducibility while steering clear of both underfitting and overfitting.

8. The majority of the "Results and discussion" section is descriptive rather than explanatory. It fails to investigate the reasons for the significant performance differences among different gases. Why is the model's prediction almost perfect for CO but poor for N₂O (AARD = 17.07%)?

Author response: Thank you for your valuable comment. The related discussion has been amended accordingly in section 3.3. Please see the highlighted section (first paragraph) on pages 36–37.

The differences in predictive performance across gases can be attributed to the inherent thermodynamic behaviour and statistical properties of each dataset, such as the number of data. For CO, the solubility patterns demonstrate consistent trends and almost linear monotonic trends concerning temperature and pressure for most ionic liquids (ILs), with the experimental value range being quite narrow and in low amounts, as can be seen in Table S5. This regularity in the CO dataset enables the GP model to capture its characteristics with near-perfect precision. In contrast, N₂O exhibits more intricate data and as Table S3 indicates that N₂O mole fractions range from 0.001–0.781, covering almost three orders of magnitude, with large variability across ionic liquids, temperature, and pressure. The N₂O dataset also features regions of very low solubility, where even minor absolute variations can lead to substantial percentage errors, thus elevating the AARD despite having reasonable RMSE and R^2 values. These factors together account for the GP model's exceptional performance with CO while demonstrating comparatively higher errors for N₂O.

9. An attempt should be made to interpret the structure of the equations. Even if a complete physical interpretation is not feasible, the authors should discuss whether these functional forms are reasonable and how they may be related to underlying physical phenomena.

Author response: Thank you for your valuable comment. The related discussion has been explained in section x. Please see the highlighted section on Page 36.

Although GP generates explicit mathematical formulas, the resulting expressions tend to be complex due to the need to account for the highly nonlinear relationship between gas solubility and factors like temperature, pressure, and the critical properties of ionic liquids. Consequently, the symbolic relationships produced by GP do not translate easily into straightforward, interpretable physicochemical principles. Instead, their significance lies in offering clear, equation-based predictions while still demonstrating physically meaningful behavior. This is illustrated in Figure 2, where the GP model accurately captures the essential thermodynamic patterns. As shown in Figure 2, the gas solubility in ILs was increased by increasing pressure and decreasing temperature. However, there is an exception for CO solubility in [BMIM][MeSO₄][154], which was increased by increasing temperature, which has been mentioned as a rare phenomenon [154]. Thus, although the GP equations do not provide new mechanistic insights into the interactions between gas and ionic liquids, they effectively encapsulate and reflect the anticipated physical trends across a diverse range of systems with great precision.

10. Two different equations are both labeled as Equation (10).

Author response: Thank you for your valuable comment. The equation numbers have been modified now. Please see the highlighted sections on Pages 33 and 34.

11. In the Conclusions section, the sentence "by considering the parameters of ω , T_c , P_c , P , ω , and T " contains a redundant ω .

Author response: Thank you for your valuable comment. It has been modified. Please see the highlighted sections on Page 47.

By considering the parameters of ω , T_c , P_c , and M_w of ILs, as well as P and T of the process.

Reviewer #2

This research summarizes the applications of machine learning (ML) in predicting gas solubility in ionic liquids (ILs) and develops a new model of genetic programming (GP). The work involves substantial effort and a comprehensive dataset, with the developed GP model and Excel tool demonstrating considerable practical appeal. My comments are shown as bellows:

(1) The authors claim that one of their primary contributions is providing a "simple" mathematical equation. However, is such a formula physically and chemically interpretable? Compared to a "black-box" neural network with comparable performance, where exactly does

this "simplicity" lie? Please elucidate the genuine merits of the proposed formula compared to existing models (such as MLP and SVM).

Author response: Thank you for your valuable comment. The related discussion has been responded to the section 3.2. Please see the highlighted section on Page 30.

While the mathematical intricacies of the correlations generated by GP are complicated, they present several advantages that black-box models fail to deliver. To begin with, the explicit equation enables users to calculate gas solubility directly without the need for training datasets, specialized machine learning software, or model retraining, thus making the correlation easily portable for use in industrial applications or offline scenarios (such as spreadsheets, process simulators, and numerical codes). Additionally, the functional format offers full transparency, allowing users to examine how temperature, pressure, and IL characteristics influence the model, which is not achievable with MLP-ANN or LSSVM frameworks. Moreover, the GP equation can be integrated into design calculations or utilized within process optimization frameworks as a closed-form function, a capability that is largely not possible with most black-box methodologies. Consequently, despite its mathematical complexity, the GP correlation provides greater interpretability, reproducibility, and practical applicability compared to traditional black-box models.

(2) The manuscript mentions that the dataset was divided into training and test sets; however, the methodology for this partition is not specified. It is crucial to clarify whether the split was random or based on specific criteria, such as ionic liquid type, temperature, or pressure ranges. The chosen strategy profoundly impacts the evaluation of the model's generalization capability. An inappropriate partition may result in the model merely memorizing the training data rather than learning the underlying patterns. Therefore, a detailed explanation of the data splitting strategy is recommended.

Author response: Thank you for your comment. The related information has been addressed in section 3.1. Please see the highlighted section on Page 27.

The parameters M_w , T_c , P_c , and ω were chosen because they reflect essential thermodynamic values that influence how gases dissolve in ILs. M_w , T_c , P_c , and ω are crucial for the ionic liquid's capacity to solvate gases. These characteristics are commonly utilized in thermodynamic models (such as PC-SAFT [133]) and in QSPR methods for forecasting gas solubility. Temperature and pressure serve as the main factors driving gas dissolution equilibrium. In addition, Numerous earlier studies utilized these parameters and demonstrated satisfactory outcomes [34, 100, 116, 125].

The full dataset, comprising 3209 experimental points, was partitioned into training (75%) and testing (25%) subsets via a random split. This method guarantees that the distributions of temperature, pressure, and IL types are proportionately maintained in both subsets. There was no grouping or selection conducted based on specific IL families, temperature ranges, or pressure intervals; rather, every sample had an equal chance of being allocated to either subset, which helps to prevent bias towards specific gas–IL combinations. The similarity in the R^2 , RMSE, and AARD values between the training and testing subsets indicates that the GP model captured underlying patterns rather than simply memorizing the training data.

(3) Page 30: "a comparison of the AARD% of GP model by heat map is presented in Table 7 for different Gas-IL systems" Should Table 7 here be changed to Figure 3? Please provide further clarification on the AARD% heat map for the Gas-IL system.

Author response: Thank you for your comment. The related information has been addressed in section 3.3. Please see the highlighted last paragraph in Section 3.3 on Page 37.

The heat map displayed in Fig. 3 illustrates the AARD% values for each Gas-IL system, facilitating a visual assessment of the prediction challenges associated with different gases. Darker shades indicate greater deviations, while lighter shades signify more precise predictions. The heat map distinctly indicates that gases like CO, H₂S and SO₂ generally show the least deviations across most ILs. Conversely, CO₂ reveals higher AARD% values for various ILs, indicative of their wider solubility ranges.

(4) What is the basis for choosing Mw, Pc, et al. as input descriptors when establishing the GP model? Is it rigorous to use the same set of descriptors to capture the physical and chemical mechanisms of the dissolution of various gases in ILs? It is suggested to add the evaluation of the feature importance of input descriptors, which not only enhances the credibility of the model but also provides valuable insights for understanding the dissolution mechanism.

Author response: Thank you for your comment. The reason for choosing input descriptors to capture the physical and chemical mechanisms of the dissolution of various gases in ILs has been addressed in the highlighted text in the first paragraph in section 3.1.

The parameters Mw, Tc, Pc, and ω were chosen because they reflect essential thermodynamic values that influence how gases dissolve in ILs. Mw, Tc, Pc, and ω are crucial for the ionic liquid's capacity to solvate gases. These characteristics are commonly utilized in thermodynamic models (such as PC-SAFT [133]) and in QSPR methods for forecasting gas solubility. Temperature and pressure serve as the main factors driving gas dissolution equilibrium. In addition, Numerous earlier studies utilized these parameters and demonstrated satisfactory outcomes [34, 100, 116, 125].

Section 3.4 has been added for the evaluation of the feature importance of input descriptors.

(5) The current assessment relying only on a train-test split is insufficient. The model is likely to have merely learned to interpolate the known data of ILs and gases within the known temperature and pressure range, but has failed to prove its ability to extrapolate and predict new systems. It is suggested to supplement the proof of the generalization ability of the GP model.

We appreciate the reviewer's valuable comment. In response, we have conducted additional validation analyses to demonstrate the generalization ability of the GP model. The results have been described in Pages 37 and 38:

To tackle the issue of model generalization beyond a straightforward train-test division, the ability of the GP model to extrapolate was assessed using independent experimental data

sourced from existing literature that was not utilized during any phase of model creation and the training data. The findings regarding extrapolation are presented in Table S8. The GP model showed acceptable predictive capability on these previously unseen datasets, showing acceptable statistical accuracy. This indicates that the model captures the fundamental physicochemical trends that influence gas solubility in ILs, rather than simply interpolating the training data. It should be noted that in extrapolation, an outlier detection procedure was applied to the external datasets to ensure robustness. Data points identified as inconsistent or erroneous in the original literature were excluded in accordance with established practices in machine learning-based thermophysical modeling. The complete extrapolation datasets are available in the Supplementary Excel file to ensure transparency. It is important to mention that we could only find two independent experimental data points related to CO available in the literature. Due to this very limited dataset, conducting a statistically valid validation of the model was not possible, and traditional performance metrics (such as R^2) cannot be considered reliable in these circumstances. As a result, the typically low R^2 value noted for CO does not indicate inadequate model performance but instead highlights the lack of sufficient experimental data. This constraint is clearly recognized in the manuscript.

(6) What are the specific advantages of the proposed GP model compared to traditional physical models (e.g., COSMO-based, UNIFAC, NRTL and PC-SAFT etc.)? Please the authors added the detailed comparison with the above-mentioned physical models in main text.

Author response: Thank you for your comment. Detailed discussions have been added accordingly. Please see the highlighted sentences in sections 2.2.1, 2.2.2 and 2.3.

Review and Development of an Easy-to-Use Machine Learning Model for Pollutant Gas Solubility in Green Solvent of Ionic Liquids

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Abstract

The increasing release of greenhouse gases (GHGs) like CO₂, CH₄, N₂O, and industrial contaminants (indirect GHGs) such as SO₂ and H₂S has prompted significant global worries due to their role in climate change, air pollution, and harm to the environment. Ionic liquids (ILs) as green solvents have emerged as promising alternatives to traditional solvents because of their minimal volatility, high thermal stability, and adjustable physicochemical characteristics. Yet, limited gas solubility data in ILs are hindering their applications in carbon capture and air pollution control. Machine learning (ML) is a powerful tool for modeling and simulating the solubility of polluting gases in ILs. This research aims to critically review recent progress in ML modeling of pollutant gas removal by ILs. More importantly, a new ML model of genetic programming (GP) was developed to generate a simple and accurate mathematical equation to predict the solubility of SO₂, CH₄, N₂O, CO, H₂S and CO₂ in ILs, using a large dataset (3209) for different gas-IL systems. Using temperature, pressure, and structural related parameters of ILs and gases as input parameters, the model achieved a high accuracy ($R^2 > 0.97$). Finally, a simple Excel method for calculating gas solubility has been created for prediction and modeling purposes.

Keywords: Genetic programming; Global warming; Ionic liquids; Machine learning; Pollutant gases

Introduction

The global warming from the emission of greenhouse gases (GHGs) has raised serious concerns in recent decades [1]. Some GHGs, including methane (CH_4) and carbon dioxide (CO_2), have direct radiative effects [2]. Industrial operations emit significant quantities of H_2S and CO_2 into the air [3]. On the other hand, indirect radiative effects can be attributed to some GHGs (e.g., SO_2) [4]. N_2O is not a major contributor to global warming (around 6%), but is one of the GHGs identified for reduction by the United Nations Framework on Climate Change (UNFCCC) [5].

CO_2 is the dominating source of GHGs, which requires robust technologies for control. Various techniques have been proposed to treat and manage the emission of CO_2 from industrial flue gases [6, 7], including compression, membrane technology, conversion, solids adsorption, liquid solvent absorption, and biological methods [8-10]. Notably, all of these techniques exploit a functional material or a high CO_2 -affinity solvent [11, 12]. The utility sector, especially the post-combustion flue gas treatment, involves the use of absorption-based technologies that employ a CO_2 absorbent (e.g., aqueous amines) [13, 14]. However, high temperatures are required for the desorption of CO_2 from absorbing fluid before reuse, hence CO_2 capture with aqueous amines is an energy-intensive process [15].

Ionic liquids (ILs) are defined as salts that have a melting point below the ambient temperature. The structure of ILs can be designed by changing their constituents (i.e., cations and anions), leading to the preparation of different ILs with pre-determined features. ILs have been applied in membrane processes, gas separation, electrochemistry, organic chemistry, lubrication, 3D printing technology, biotechnology, quenching and catalysis, and extraction [16-19]. Since ILs have special characteristics that are suitable for CO_2 capture (e.g., negligible volatility, tunable reaction enthalpy with CO_2 , high CO_2 solubility, and wide liquidus range), they have been promoted as alternative solvents to aqueous amines [20, 21].

Several reports have studied the solubility of gases (e.g., CH_4 , SO_2 , N_2O) in ILs [22-24] and assessed CO_2 capture by ILs-based solvents [25-27]. However, the experimental measurement can be a

complex, costly, and arduous process. Since there is a rapid growth in the number of new ILs categories, there will be a scarcity of experimental data related to new ILs. Accordingly, predictive methods have been developed to calculate the CO₂-IL systems' phase behavior. In addition, the solubility of various gases in ILs has been modeled applying thermodynamic laws, correlations, and innovative approaches [28-30]. For instance, the phase behavior of CO₂/IL mixtures has been predicted by applying the thermodynamic-based models (e.g., Generalized Corresponding States-EOS, Redlich-Kwong, temperature-dependent Perturbed-Chain Statistical Associating Fluid Theory (tPC-PSAFT), soft-SAFT, Perturbed-Hard-Sphere Theory (PHST), Statistical Wertheim-Tanh Chain Fluid (SWCF), and Peng-Robinson (PR) [28, 30, 31]. However, thermodynamic models have low accuracy in certain cases.

There is a growing interest in machine learning (ML) based methods owing to their ability to model linear and non-linear systems, as well as flexibility and simplicity [29, 32-34]. There has been a growing interest in predicting pollutant gas capture, especially CO₂, by various ML models [35-45]. So far, no research has proposed a simple model to predict gas solubility in ILs. Therefore, this research aims to review recent progresses on the solubility of polluting gases in ILs and the ML modeling of gas capture by ILs; to propose Genetic programming (GP) model encompassing a general correlation based on Temperature (T), Pressure (P), molecular weight of IL (M_w), critical temperature of IL (T_c), critical pressure of IL (P_c) and accentric factor (ω); and to predict the solubility of CO, H₂S, N₂O, CH₄, SO₂, and CO₂ in ILs at different temperature and pressure.

2. Theory

2.1. ILs and their gas dissolution

ILs have been studied for over a century, with the first reports dating back to 1914 [46], followed by the development of air and water-stable ILs in 1992 [47-49]. Since then, numerous books have investigated their applications in areas such as chemical production, bioprocessing, electrochemistry, medical, catalysts and separation processes [50-60]. ILs continue to attract significant attention due

to their minimal vapor pressure, high chemical and thermal stability, and the versatility offered by diverse cation and anion combinations [61]. In particular, their ability to capture and separate CO₂ makes them promising candidates for mitigating greenhouse gas emissions and reducing environment impacts [62]. To understand the application of ML in gas-IL systems, it is essential to review the fundamental properties of ILs and existing predictive methods.

2.2. Methods for gas solubility data

The experimental methods for determining gas solubility in ILs involve physical approaches (e.g., gravimetric, synthetic/stoichiometric) and analytical techniques (e.g., chromatographic and spectroscopic) [63]. While several of the latest studies have incorporated analytical methods [64-68], physical methods remain most popular. More details are provided in the Supplementary material.

There are various methods for estimating gas solubility in ILs. These methods are equation of states (EOSs) and computational modelling of gas solubility in ILs, which includes approaches that blend quantum chemical calculations with statistical thermodynamics and correlative and empirical methods.

2.2.1. Equations of state (EOS)

EOSs are mathematical formulations that define the relationships among different thermodynamic properties in a system when it is in a state of equilibrium. EOSs are empirical formulas that explain the unique characteristics of a system and can be derived from either experimental data or molecular concepts. Using approaches such as the Peng-Robinson (PR) EOS [69] and the Soave-Redlich-Kwong (SRK) EOS [70], it is possible to estimate the solubility of acidic gases in different ILs. As a fundamental theory grounded in natural principles, thermodynamics provides a foundation for creating EOSs for different materials. These equations are established based on experimental data that precisely determine a system's thermodynamic properties across a range of values. However, an EOS is only as accurate as the experiments that inform its development, and its validity is limited to

the measured range of values. If a system operates outside this range, a new EOS may be established [71].

To simulate the interactions between gas and IL systems, activity coefficient models and EOS methods have been widely employed. Non-random two-liquid (NRTL) model, as a model of activity coefficient, has been employed to analyze experimental data related to fluorinated gases [72-75], CO₂ [76], NH₃ [77], and hydrocarbons in ILs. Another activity coefficient model, known as Universal Quasi-chemical Functional-Group Activity Coefficients (UNIFAC), utilizes the interactions of functional groups in molecules to compute activity coefficients. UNIFAC was employed to investigate NH₃ and H₂O in ILs [77], but its applicability may depend on the manner in which functional groups are specified within the ILs. Variations of Universal Quasi-chemical Functional-Group Activity Coefficients, like UNIFAC-Lei [78-80] and UNIFAC-IL [81], have been used to model solubilities, screen ILs for various gases, and assess their potential for CO₂ capture [82]. Nevertheless, the activity coefficient models exhibit shortcomings, particularly when operating at elevated temperatures, especially in the proximity of or surpassing the gas species' critical temperature (T_c). In order to remove these limitations, using an EOS is a more dependable approach for predicting the overall gas phase characteristics in ILs. Many EOS models depend on critical parameters. However, calculating these values for ILs can be challenging, as their estimated T_c values often surpass established decomposition temperatures. The determination of pseudo-critical properties for ILs has been found to exert negligible influence on calculations related to phase behavior [83]. The Statistical Associating Fluid Theory (SAFT) EOS model is a forward-looking approach that considers aspects like molecular dimensions, shape, and the presence of hydrogen bonding. Numerous variations of SAFT are available, with Soft-SAFT being one of them. Soft-SAFT utilizes a Leonard-Jones fluid as a reference to characterize atmospheric gases within fluorinated ILs [84, 85] and refrigerant gases in ILs [86]. The other offshoot is the perturbed chain SAFT (PC-SAFT), which is employed to model acidic gases [85] and CO₂ in ionic liquids. For ion-specific electrolyte perturbed-chain SAFT (ePC-SAFT), it forecasts CO₂ solubility in a wide array of ILs [87]. A recent

adaptation known as critical point-based perturbed chain theory (CP-PC-SAFT) utilizes correlation according to molecular weight to predict vapor-liquid equilibrium (VLE) for different gases in imidazolium ILs [88]. Given that IL molecules consist of asymmetrically neutral ion pairs, resulting in a significant dipole moment resulting from the distribution of charge across these pairs, and considering that CO₂ molecules possess a quadrupole moment, it becomes imperative to apply thermodynamic relationships that account for these strong interactions [89].

Additionally, it is important to factor in the influence of high pressures typically encountered in supercritical processes. Existing thermodynamic models have been primarily tailored to specific systems, primarily due to these considerations, rather than being designed for a range of systems, which is the focus of this research. Hence, it is imperative to extend comprehensive approaches capable of forecasting phase behavior in diverse IL systems, which are now being explored as alternatives to conventional industrial solvents [90].

2.2.2. Computational modeling of gas solubility in ILs

In the realm of hybrid quantum chemical/statistical thermodynamics methods, as quantum chemical approaches yield highly exact energy interaction between individual molecules, they face challenges in directly computing solubilities. This is due to the complexity of the effects of solvation, which encompasses interactions among numerous molecules and makes precise quantum calculations computationally expensive. Furthermore, an accurate consideration of entropic impacts requires the exploration of diverse arrangements of solute and solvent molecules, a task that is economically unfeasible with straightforward quantum techniques [91]. Some investigators have ventured into the utilization of compact cluster models to reveal solvation trends [91], yet these generally furnish qualitative outcomes and lack the capacity to yield macroscopic solubility projections. Consequently, solvation models are frequently amalgamated with quantum chemical approaches for solvation prediction [92].

A well-known example of this approach is the Conductor-like Screening Model (COSMO) and its modified versions. These models were applied for screening ionic liquids for special applications, such as the separation of gases [93]. COSMO-RS [94], for instance, was used to determine the CH₄ solubility in ionic liquids and gain insights into interactions at the molecular level. While it couldn't make precise quantitative predictions, it could assess the impact of various IL characteristics, like anions and cations. COSMO-RS and variations have also been used for predicting the solubility of ethane [95] and SO₂ [96] in ILs. Additionally, COSMO-SAC was employed for modeling vapor-liquid equilibrium of CO₂ [97, 98] and H₂S in ILs, although it has been recognized as unsuitable for modeling N₂ and oxygen in IL [99].

2.3. Machine Learning, optimization and predictive modelling

Machine learning (ML) has emerged as a powerful tool for modelling gas solubility in ILs, which enhances model accuracy and robustness. The development of predictive models typically involves structured stages of data preprocessing, training, and testing, with workflow examples illustrated in Figure S1. A detailed description of ML algorithms, optimization approaches, and model development procedures is provided in the Supplementary Materials. In this study GP model is used for modeling. The advantages are as follows:

(i) The GP model relies solely on easily accessible descriptors (Mw, Tc, Pc, ω , T, and P), making it suitable for use even when detailed structural or quantum-chemical data is not available. (ii) GP autonomously identifies both the functional form and parameter values directly from the dataset, allowing it to capture intricate nonlinear effects and interactions between IL properties and operational conditions that might be challenging to represent in predefined analytical forms like EOS or activity-coefficient models. (iii) Unlike black-box ML models, GP produces a clear algebraic equation, providing interpretability and facilitating straightforward implementation in engineering calculations or process simulators without the need for iterative EOS solving. (iv) After training, the GP model demonstrates low computational demand and enables a unified analysis of multiple gases

and IL families using the same set of descriptors, whereas most physical models necessitate distinct parameterization for each gas-IL pairing. These benefits render GP a versatile and effective supplementary method for screening or predicting gas solubility in various IL systems.

2.4. Machine learning modeling of Gas capture by ILs

2.4.1. Machine learning modeling of CO₂ capture by ILs

One of the most important GHGs, which is the primary contributor to global warming and is gaining growing attention, is CO₂. Table 1 summarizes the key studies on CO₂ solubility prediction using various ML models, presenting performance outcomes across diverse datasets and IL systems. An ANN model was employed to model the solubility of supercritical CO₂ in a selection of 24 widely used ILs. Impressively, this model exhibited robust predictive capabilities, achieving an R² of 0.993 and maintaining an average absolute deviation of 3.6% when applied to a substantial dataset of 1128 data points [90]. Another study harnessed Artificial neural networks (ANNs) as a primary tool, using a dataset of 2930 CO₂ solubility data, and the outcomes underscored the exceptional proficiency of the ANN model in establishing highly accurate correlations for acid gas solubility in ILs [33]. Feng et al. [100] used twelve distinct ML algorithms to estimate the solubility of CO₂ and H₂S in ILs, based on 4107 data points for CO₂ in 17 different ILs and 549 data points for H₂S in 10 distinct ILs. The collective findings highlighted the dominance of ensemble algorithms. In the case of the IL-CO₂ system, the stacking approach emerged as the most accurate model (R² = 0.969). Other methods, including Random Forest, Bagging, AdaBoost, and Gradient Boosting, delivered similar enhanced solubility estimations, with R² values of 0.959, 0.954, 0.953, and 0.948, respectively. Deng et al. [101] introduced theoretical prediction models for CO₂ solubility in ionic liquids by blending the critical properties of ILs with deep learning models, achieving R² between experimental and predicted values of 0.988 in CP-RNN, 0.999 in CP-CNN, and 0.984 in CP-DNN.

In a separate investigation, ANN and Support Vector Machine (SVM) were applied to a dataset encompassing 10116 CO₂ solubility data points across 124 diverse ILs, with temperatures ranging

from 243.2 K to 453.15 K and pressures from 0.00798 bar to 499.9 bar. The ANN-GC (group contribution) model achieved an R^2 of 0.9836, while the SVM-GC model reached 0.9783 [102]. Tatar et al. [34] employed Radial Basis Function (RBF), Multilayer Perceptron (MLP), Least Squares Support Vector Machine (LSSVM) and committee machine intelligent system (CMIS) to predict CO_2 solubility in 14 different ILs. The utilization of GA facilitated the determination of CMIS coefficients, leading to impressive results with a high R^2 value of 0.9992 for CMIS in predicting CO_2 solubility in ILs. In another investigation, a dataset of 728 experimental data points was employed for the implementation of MLP-ANN and Adaptive Neuro-Fuzzy Inference System (ANFIS) models. The MLP-ANN model showcased a notable R^2 value of 0.9972 and a low MSE of 0.000133. In contrast, the ANFIS model achieved R^2 of 0.9336 and a mean squared error (MSE) of 0.002942. Conversely, the Peng-Robinson-EoS (PR-EoS) model resulted in an R^2 of 0.7323 and an MSE of 0.002702, while the Soave-Redlich-Kwong (SRK-EoS) model displayed an R^2 of 0.6989 and an MSE of 0.005578. As a result, the current study underscored the superior performance of MLP-ANN as an exceptionally accurate model for estimating CO_2 solubility in ILs [103]. In another study, a correlation was developed to predict the solubility of CO_2 in 27 commonly utilized ILs. Even when the binary interaction coefficients of PR and SRK were fine-tuned using experimental data, the proposed model consistently outperformed them in terms of accuracy. Specifically, the Average absolute relative deviation (AARD%) values were 4.9% for the proposed model, 43.5% for PR, and 44.3% for SRK when considering a total of 3073 data points [104].

A three-layer feed-forward ANN was developed and optimized using a genetic algorithm to predict the solubility of CO_2 in ethanol–[EMIM][Tf2N] IL mixtures, with close agreement with the experimental data [105]. Furthermore, the absorption of CO_2 in polyionic liquids was modeled using RBF and LSSVM models. The RBF model demonstrated an AARD% of 2.68 and an R^2 of 0.996. On the other hand, the LSSVM model showcased an AARD% of 0.271 and approached a perfect R^2 value of 1.00 [106]. Furthermore, an extensive database comprising 5368 data points from 67 diverse ILs was utilized. The results obtained through the employment of the LSSVM approach exhibited a higher

degree of accuracy when compared to the RBF-ANN, MLP-ANN, and ANFIS models. The models achieved high performance as shown by R^2 and MSE values: LSSVM (0.9942 & 0.00035), ANFIS (0.9135 & 0.004883), MLP-ANN (0.9135 & 0.004883), and RBF-ANN (0.9135 & 0.004883) [32].

In addition, the equilibrium absorption of CO₂ in various ILs was modeled using a decision tree learning algorithm known as Classification and Regression Tree (CART). This model leveraged data from over 5360 experimental datasets on CO₂ equilibrium solubility in 67 different ILs across a range of temperatures and pressures. The model yielded a %AARD of 0.04% and delivered notably superior estimations when compared to previously published RBF-ANN, MLP-ANN, ANFIS, and LSSVM models [107]. Furthermore, an investigation delved into the utilization of the Group Method of Data Handling (GMDH) intelligent method as a computational strategy for the prediction of solubility of CO₂ in a variety of ILs at temperatures both below and above 324 K, using a dataset of 4726 data points in 60 distinct ILs. The model showcased the potential as an effective predictor for the estimation of CO₂ solubility in ILs, offering prospective applications across various industries [108].

Mesbah et al. estimated CO₂ solubility in 24 distinct ILs across an extensive range of temperatures (283.15–373.35 K) and pressures (99.8–85800 kPa) using GMDH correlation, RBF, and empirical correlation. These models collectively excelled in providing exceptionally accurate estimations corresponding to experimental values. To ensure the robustness of the database, a diagnostic test for statistical outliers was thoughtfully executed [109]. Another study used the amalgamation of GA-MLR, Genetic Algorithm-Multiple Linear Regression (GA-MLR), and regression models were employed to formulate a nonlinear quantitative-structural property relationship (QSPR) model for the prediction of CO₂ solubility in ILs. This innovative model hinged on the utilization of three pivotal input descriptors: the number of single bonds (N_{sb}), Average bonding information content (order 1) ($\overline{BIC}^1$) and Image of the Onsager-Kirkwood solvation energy SE_{ok} . The nonlinear model (LS-SVM) delivered highly reliable predictions, with an R^2 of 0.962 for the entire dataset, while the linear MLR model achieved an R^2 of 0.876 [110].

In addition, LSSVM, decision trees (DT), RF, and Multilinear Regression (MLR) were employed to estimate CO₂ solubility in a diverse range of ILs, based on thermodynamic properties and QSPR models. The research used two datasets related to CO₂ solubility, one relying on critical properties, molecular weight, and the acentric factor of pure ILs as input data, and the second dataset incorporating structural descriptors of cations and anions as input parameters. The study revealed a strong agreement between the predicted values and experimental measurements, especially when employing RF and DT techniques developed based on descriptors and thermodynamic properties. Models based on structural descriptors outperformed those built on critical properties, and feature selection methods were applied to pinpoint the most effective cation and anion descriptors [111]. Furthermore, Particle swarm optimization (PSO)-RBF, MLP-ANN, Coupled simulated annealing (CSA)-LSSVM, and Hybrid-ANFIS models were harnessed to forecast the solubility of CO₂ in ILs, using a dataset of 2930 data points involving 39 CO₂-IL systems. The outcomes revealed that the developed models yielded accurate predictions with reasonable margins of error. Notably, the CSA-LSSVM model displayed enhanced predictive capabilities for CO₂-IL. Additionally, it was evident that the predictions from these models outperformed the thermodynamic model, demonstrating the feasibility of correlating CO₂ solubility in ILs with their structural parameters [112]. As shown in Table 1, the CART algorithm delivered perfect predictions ($R^2 = 1.000$, AARD = 0.04%) [107], highlighting the capacity of ML techniques to surpass traditional approaches for CO₂ solubility estimation

2.4.2 H₂S solubility in ILs

Table 2 presents a comprehensive comparison of ML models for H₂S solubility prediction. While CO₂ capture has received extensive attention (Table 1), hydrogen sulfide (H₂S) removal presents different challenges due to its toxicity and corrosive nature.

Notably, Extreme gradient boosting (XGBoost) achieved exceptional accuracy ($R^2 = 0.9998$) with minimal error (AARD = 1.14%), suggesting that H₂S prediction may be more amenable to ML

modeling than CO₂ due to stronger specific interactions. Hosseini et al. utilized an MLP-ANN model to forecast the solubility of H₂S in ILs, using a dataset of 1243 data points from 33 ILs, with an impressive R² value of 0.9976 [113]. Mousavi et al. utilized 1516 real data points to forecast H₂S solubility in 37 ILs. They developed four models, including the deep belief network (DBN), GMDH, Gaussian Process, and XGBoost, to compute H₂S solubility in ILs. Among these models, XGBoost was found to be the most precise and reliable [114]. In another research endeavour, 12 distinct ML algorithms were enlisted to estimate H₂S solubility in various ILs. Notably, the Voting algorithm outperformed others in the IL-H₂S system, yielding MSE and R² values of 0.002 and 0.934, respectively [100]. In a separate study, diverse deep learning techniques were harnessed to predict H₂S solubility in ILs. These models encompassed CNN, Recurrent Neural Network (RNN), Deep Belief Network (DBN), and DJINN. Utilizing a substantial databank comprising 1516 data points concerning H₂S solubility in 37 different ILs, the outcomes revealed that the CNN model offered more precise, expedient, adaptable, and cost-effective estimations for H₂S solubility in ILs, achieving an impressive R² of 0.99 [115].

Abdi et al. conducted a comparative analysis of LSSVM, ANFIS, MLP-ANN, Cascade Feed-Forward Neural Network (CFFNN), Generalized regression neural network (GRNN), and RBF, to predict H₂S solubility in ILs, using an experimental databank of 792 datasets. The results demonstrated that LSSVM accurately predicted H₂S solubility in ILs, with R², MSE, and AARD values of 0.9980, 0.00012, and 4.03, respectively [116]. PSO-RBF, MLP-NN, CSA-LSSVM, and Hybrid-ANFIS were successfully deployed to predict the solubility of H₂S in ILs, using a dataset of 664 data points encompassing 14 H₂S-IL systems. Additionally, the model predictions were assessed against those generated by a thermodynamic model grounded on the PR EoS. The outcomes underscore that the developed models offer precise predictions with acceptable margins of error. Particularly noteworthy is the CSA-LSSVM model, which excels in projecting H₂S solubility in ILs, outperforming the thermodynamic model's projections. This study underscores the potential for establishing a correlation between H₂S solubility in ILs and the structural attributes of ILs [112].

Baghban et al. estimated the solubility of supercritical H₂S in ILs utilizing the LSSVM method optimized via GA. They also incorporated a group contribution approach to reduce the model's dependence on experimental data. The model yielded root mean square error (RMSE) and R² values of 0.0122 and 0.9941, respectively [117]. Another investigation was conducted to assess the efficacy of a state-of-the-art soft computing technique known as gene expression programming (GEP) in estimating H₂S solubility within ILs. A dataset comprising 465 experimental data points derived from 11 different ILs was employed to devise a comprehensive correlation. The modeling outcomes yielded an R² of 0.9902 for the projected solubilities compared to the corresponding experimental values. Furthermore, the predictions generated by the GEP model were juxtaposed with the results of two well-established engineering methodologies, SRK and PR. The GEP method proved to be more precise than these widely utilized thermodynamic models [118].

Amedi et al. devised models based on MLP-ANN, ANFIS, and RBF-ANN to foresee the solubility of H₂S in a variety of ILs, using a compilation of 664 experimental data points from the literature. In predicting H₂S solubility within a ternary mixture containing CO₂ through the MLP-ANN approach, 89 experimental data points were set in motion. The MLP-ANN model yielded a commendable R² of 0.9951 and an MSE of 0.000117. In contrast, the ANFIS and RBF-ANN techniques delivered R² and MSE values of 0.901 and 0.002268, and 0.9679 and 0.000787, respectively. These results underscore the precision of the MLP-ANN model in estimating H₂S solubility in ILs in comparison to other computational intelligence models [119]. The study delved into the potential of ANNs that underwent training and optimization using BP and PSO to establish correlations for the solubility of H₂S in 11 distinct ILs. A dataset comprising 465 experimental data points, gleaned from existing literature, served as the foundation for these models. The PSO-ANN model displayed a noteworthy R² of 0.99218 and a MSE of 0.00025. In contrast, the BP-ANN model yielded an R² of 0.9515 and an MSE of 0.00335. The outcomes signify that, according to statistical criteria, the PSO training approach surpasses the BP training methodology [120].

Ahmadi et al. [121] presented the utilization of the LSSVM for the prediction of H₂S solubility in diverse ILs. Their approach involved harnessing a genetic algorithm (GA) to acquire the most suitable hyperparameter values for LSSVM, encompassing parameters like γ and σ^2 . Using 465 data samples involving 11 ILs, their approach demonstrated high predictive performance, achieving an R^2 of 0.9976 and an MSE of 0.000066. Kang et al. [122] introduced novel molecular descriptors referred to as electrostatic potential surface (SEP) to forecast H₂S solubility in ILs. They amassed 1318 experimental data points from 28 ILs and, through the amalgamation of SEP descriptors and the Extreme Learning Machine (ELM) algorithm, constructed two quantitative models (ELM1 based on isolated ions and ELM2 based on ion pairs) for the prediction of H₂S solubility. The AARD% for the comprehensive set of ELM1 and ELM2 models was 5.87% and 3.84%, respectively, underscoring the efficacy of SEP descriptors in predicting IL properties due to their rich electron-level information. Amar et al. introduced the Ionic Solubility Compound Model (ISCM) to forecast H₂S solubility in a range of ILs. The ISCM models were established through the integration of robust data-driven methodologies, including MLP and CFNN, using systematic techniques like the GMDH and GP. These models were constructed utilizing an extensive dataset comprising 1243 data points for H₂S solubility in 33 ILs. The ISCM-GP model they formulated demonstrated AARD and R^2 values of 2.3767% and 0.9990, respectively. Additionally, trend analyses confirmed that ISCM-GP predictions closely mirrored real variations in H₂S solubility in ILs [123]. Detailed information regarding ML modeling of H₂S solubility in ILs is shown in Table 2.

As indicated in Table 2, CSA-LSSVM and PSO-ANN emerged as the most accurate approaches (R^2 = 0.9999, 0.9998 and AARD = 0.13, 1.3, respectively). These results confirm that H₂S solubility can be effectively captured with ML, with performance levels far exceeding classical thermodynamic correlations.

2.4.3. N_2O , SO_2 and other gas solubility in ILs

Table 3 compiles the major ML models applied to N_2O , SO_2 and other gases (mixed gases, CH_4 , CO , Ar , O_2 , N_2 , C_2H_6) solubility prediction in ILs, s, demonstrating their predictive efficiency compared to other models.

Three distinct machine learning approaches, encompassing ANN, SVM, and LSSVM, were utilized to foresee the solubility of N_2O in a set of 25 ILs. The optimization of hyperparameters for SVM and LSSVM models was carried out using the Shuffled Complex Evolution (SCE) algorithm, while the optimal architecture for the ANN model was determined through a trial-and-error method. An evaluation of these three models unveiled that SVM outperformed the others in predicting N_2O solubility in the 25 examined ILs, achieving an R^2 value of 0.9970 [124]. In a different research endeavor, four robust models (deep belief network (DBN), Cat-Boost, ELM, XGBoost) were constructed using two distinct methodologies, one rooted in chemical structure-based information (I) and the other relying on thermodynamic properties-based data (II). Several EoSs were employed for comparative purposes. The findings demonstrated that these innovative approaches offered effective estimations of N_2O solubility in ILs, with XGBoost displaying exceptional performance in both methodologies (R^2 of 0.9999 and 0.9998 for Methods I and II). Significantly, the ML models outperformed the EoSs [125].

Amar et al. introduced three advanced soft-computing methods, specifically the Cascaded Forward Neural Network (CFNN), Radial Basis Function Neural Network (RBFNN), and GEP, for the purpose of forecasting N_2O solubility in 13 ILs. The outcomes illustrated that newly implemented models exhibited the capability to predict N_2O solubility in ILs with a notably high level of precision. Among these models, the CFNN model, optimized using the Levenberg-Marquardt (LM) algorithm, emerged as the most effective predictive framework, achieving an impressive R^2 value of 0.9994 [126]. Further insights into the ML modeling of N_2O solubility in ILs are shown in Table 3.

Predicting the solubility of SO_2 in various ILs involved a fusion of LSSVM and the GCM. A dataset, which comprised 232 data points concerning SO_2 solubility in different ILs, was curated for the

formulation of the LSSVM model. The model's parameters, fine-tuned through a GA. The suggested hybrid LSSVM model delivered impressive results, boasting an R^2 value of 0.9978 and an AARD% of 1.42% [127]. Bahmani et al. harnessed the power of ANN to provide estimations for the solubility of SO_2 in eight distinct ionic liquids. The study's outcomes underscore the proficiency of the ANN model in gauging SO_2 solubility in ionic liquids, yielding an R^2 value of 0.9861 and an AARD% of 2.82% [128]. Further insights into the ML modeling of SO_2 solubility in ILs are presented in Table 3.

Dashti et al. assessed numerous ILs for the purpose of gas separation, with a specific emphasis on the solubility of natural gas components, encompassing CH_4 and CO_2 in 11 distinct ILs. To establish connections between input parameters and gas solubility, they harnessed two ML methodologies, namely CSA-LSSVM and PSO-ANFIS. The results underscored the superior performance of the LSSVM algorithm, yielding an R^2 value of approximately 0.9846 and an AARD% of roughly 3.0410. In contrast, the PSO-ANFIS method achieved an R^2 of 0.9397 with an AARD% of 14.1286 [129].

In a separate investigation, a modeling approach based on ANN and molecular properties was introduced to predict the low-pressure solubility of seven gases (CO_2 , CO , Ar , O_2 , N_2 , CH_4 , and C_2H_6) in 1-butyl-3-methylimidazolium tetrafluoroborate. This method effectively estimated gas solubility by leveraging available solubility data for other gases [130]. Another study harnessed the LSSVM algorithm as a dependable tool to forecast the solubility of CO_2 -rich gaseous mixtures containing H_2S , SO_2 , CH_4 , and N_2O in various ILs. GA was utilized to determine the LSSVM hyperparameters (σ^2 and γ). The findings demonstrated that the LSSVM model's predictions were in excellent alignment with actual data points. In comparison to MLP-ANN, RBF-ANN, and ANFIS, the LSSVM model showcased superior statistical performance, yielding R^2 values of 0.9999, 0.9956, 0.9993, and 0.9628, respectively [131]. Amirkhani et al. assessed the absorption of CO_2 , CH_4 , H_2S , N_2O , SO_2 , and CO in ILs. Their study utilized a combination of MLP-ANN, Hybrid-ANFIS, PSO-ANFIS, and CSA-LSSVM methodologies, drawing upon a dataset that incorporated 3060 data points from 72 IL-gas combinations spanning 40 distinct IL varieties. Notably, the CSA-LSSVM model delivered the

most accurate predictions, demonstrating an AARD of 8.7%. These results underscore the strong connection between gas solubility in ILs and the structural attributes of the ILs [132].

These results confirm that ML models can reliably capture solubility trends not only for CO₂ and H₂S but also for a broader spectrum of industrially relevant gases in ILs. Although significant progress has been made in applying ML to model individual gas-IL systems (Table 3), relatively few studies have focused on developing models capable of handling multiple gases. This research seeks to address this gap through the comprehensive methodology outlined below.

Table 1. Modeling and simulation of CO₂ solubility in different ILs using AI methods

No.	Gas	No. of IL	Inputs of the corresponding model	Type of model	Dataset	R ²	AARD%	RMSE	Ref.
1	CO ₂	17	T, P, Mw, T _c , P _c	Linear Regression	Total	0.615	-	0.111	[100]
				Random Forest		0.959	-	0.026	
				KNN		0.925	-	0.033	
				SVM		0.037	-	0.163	
				Decision Tree		0.911	-	0.044	
				Bayesian ARD		-0.125	-	0.180	
				Gradient Boosting		0.948	-	0.029	
				Bagging		0.954	-	0.03	
				AdaBoost		0.953	-	0.029	
				NN-MLP		0.696	-	0.085	
				Voting		0.927	-	0.046	
				Stacking		0.969	-	0.0250	
2	CO ₂	13	T, P, Mw, T _c , P _c , ω	DNN	Total	0.984	-	0.757	[101]
				CNN		0.999	-	0.206	
				XGBoost		0.981	-	0.175	
				RNN		0.988	-	0.651	
3	CO ₂	124	T, P, 51 different groups present in the IL molecule	ANN-GC	Total	0.9836	-	-	[102]
				SVM-GC		0.9783	-	-	
4	CO ₂	14	T, P, T _c , P _c , ω , Mw	MLP	Total	0.9992	9.470	0.0051	[34]

				RBF		0.9984	4.707	0.0085	
				LSSVM		0.9976	9.359	0.0104	
				CMIS		0.9992	7.035	0.0059	
				MLP-ANN		0.9972	-	0.01152	
5	CO ₂	14	T, P, T _c , P _c , ω , Mw	ANFIS	Test	0.9336	-	0.0542	[103]
				PR EoS		0.7323	-	0.0519	
				SRK EoS		0.6989	-	0.0746	
				A new correlated equation		-	4.90	-	
6	CO ₂	27	T, P	PR EoS	Total	-	17.2	-	[104]
			T, P, T _c , P _c , ω	SRK EoS		-	16.9	-	
			T, P, T _c , P _c , ω	PR EoS ($k_{ij}=0$)		-	43.5	-	
			T, P, T _c , P _c , ω	SRK EoS ($k_{ij}=0$)		-	44.3	-	
7	CO ₂	24	T, P, T _c , P _c , ω	ANN	Total	0.993	3.60	0.06	[90]
8	CO ₂	1	T, P, wt. % ethanol 1, wt. % IL	ANN based on GA	Test	0.994	4.09	-	[105]
9	CO ₂	35	T, P, number of chemical structures	RBF	Total	0.996	2.67	0.829	[106]
				LSSVM		1.000	0.27	0.187	
10	CO ₂	66	T, P, T _c , P _c , ω	CART	Total	1.000	0.04	-	[107]
11	CO ₂	67	T, P, T _c , P _c , ω	LSSVM	Total	0.9942	-	0.01887	[32]
				ANFIS		0.9168	-	0.07127	
				MLP-ANN		0.9715	-	0.04169	
				RBF-ANN		0.9789	-	0.03547	

12	CO ₂	24	T, P, T _c , P _c , ω , Mw	GMDH correlation	Total	0.9650	8.46	4.2727	[109]
				RBFN		0.9942	4.23	1.7172	
				Empirical correlation		0.8860	4.25	2.4248	
13	CO ₂	46 (dataset 1) 40 (dataset 2)	T, P, Mw, T _c , P _c , ω , (dataset 1) T, P, Homo-Lumo fraction, Chi_G/D 3D, SpMax_RG, Disps (dataset 2)	DT ₁	Test	0.940	21.24	-	[111]
				LSSVM ₁		0.750	31.01	-	
				RF ₁		0.960	12.05	-	
				MLR ₁		0.550	40.48	-	
				DT ₂		0.958	20.26	-	
				LSSVM ₂		0.760	44.64	-	
				RF ₂		0.983	12.36	-	
				MLR ₂		0.282	55.36	-	
14	CO ₂	60	T, P, T _c , P _c , ω	Hybrid-GMDH	Total	0.9043	-	0.0765	[108]
15	CO ₂	21	T, P, Nsb, $\overline{\text{BIC}}$, SE _{ok}	LSSVM	Total	0.962	-	0.015	[110]
				MLR		0.877	-	0.027	
16	CO ₂	39	T, P, T _c , P _c , ω	ANN	Total	0.9947	3.970	-	[33]
				PR EoS (k _{ij} =0)		-	38.18	-	
				PR EoS (k _{ij} ≠0)		-	14.12	-	
				Empirical model		-	5.480	-	
17	CO ₂	39	T, P, Mw, T _c , P _c , ω	PSO-RBF	Total	0.9918	4.30	0.0204	[112]
				MLP-ANN		0.9872	4.33	0.0256	
				CSA-LSSVM		0.9950	1.97	0.0160	

				Hybrid-ANFIS	0.9835	3.65	0.0291	
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Table 2. Modeling and simulation of H₂S solubility in different ILs using AI methods

No.	Gas(es)	No. of IL	Inputs of corresponding model	Type of model	Dataset	R ²	AARD%	RMSE	Ref.
18	H ₂ S	37	T, P, Mw, T _c , P _c , ω, T _b	XGBoost	Total	0.9998	1.1404	0.0024	[114]
				DBN		0.9947	7.9673	0.0126	
				GP		0.7867	25.5975	0.0802	
				GMDH		0.8895	16.5551	0.0577	
19	H ₂ S	10	T, P, Mw, T _c , P _c ,	Linear Regression	-	0.874	-	0.045	[100]
				Random Forest		0.867	-	0.043	
				KNN		0.759	-	0.057	
				SVM		0.119	-	0.099	
				Decision Tree		0.785	-	0.054	
				Bayesian ARD		0.607	-	0.077	
				Gradient Boosting		0.830	-	0.048	
				Bagging		0.866	-	0.044	
				AdaBoost		0.831	-	0.047	
				NN-MLP		0.854	-	0.048	
				Voting		0.934	-	0.032	
				Stacking		0.932	-	0.033	

20	H ₂ S	37	T, P, Chemical structure of ILs	CNN	Total	0.9991	2.9251	0.0052	[115]
				DBN		0.9902	4.8636	0.0172	
				RNN		0.9960	5.5916	0.0110	
				DJINN		0.9947	6.3265	0.0126	
21	H ₂ S	15	T, P, T _c , P _c , ω	ANFIS	Total	0.9976	5.17	0.0117	[116]
				LSSVM		0.9980	4.03	0.0107	
				MLPNN		0.9977	4.58	0.0114	
				CFFNN		0.9974	5.06	0.01224	
				GRNN		0.9907	4.36	0.0231	
				RBFNN		0.9344	32.5	0.06045	
22	H ₂ S	14	T, P, T _c , P _c , ω	ANN	-	0.9987	2.070	-	[33]
				PR EoS ($k_{ij}=0$)		-	196.8	-	
				PR EoS ($k_{ij}\neq 0$)		-	8.350	-	
				Empirical model		-	5.030	-	
23	H ₂ S	14	T, P, Mw, T _c , P _c , ω	PSO-RBF	Total	0.9998	1.30	0.0024	[112]
				MLP-ANN		0.9995	1.76	0.004	
				CSA-LSSVM		0.9999	0.13	0.0006	
				Hybrid-ANFIS		0.9998	1.40	0.0027	
24	H ₂ S	27	T, P, Chemical structure of ILs	GC-LSSVM	Total	0.9957	2.74	0.0111	[117]
25	H ₂ S	11	T, P, T _c , P _c , ω	GEP	Total	0.9902	4.38	0.0165	[118]
26	H ₂ S	14	T, P, T _c , P _c , Mw	MLP-ANN	Test	0.9951	-	0.0108	[119]

				ANFIS		0.9010	-	0.0476	
				RBF-ANN		0.9679	-	0.028	
27	H ₂ S	11	T, P, T _c , P _c , ω	BP-ANN	Total	0.951	-	0.0574	[120]
				PSO-ANN		0.992	-	0.0141	
28	H ₂ S	11	T, P, T _c , P _c , ω	GA-LSSVM	Total	0.9976	-	0.0082	[121]
29	H ₂ S	33	T/ T _c , P/ P _c , T _c , P _c , ω	MLP-LMA	Total	0.9976	2.5660	0.0087	[113]
30	H ₂ S	28	T, P, electrostatic potential surface (S _{EP}	ELM ₁	Total	0.9910	5.87	0.0166	[122]
				ELM ₂		0.9940	3.84	0.013	
				MLP-LMA		0.9988	2.65	0.006	
				MLP-BR		0.9984	3.33	0.0071	
31	H ₂ S	33	T, P, T _c , P _c , ω	CFNN-LMA	Total	0.9985	3.06	0.0068	[123]
				CFNN-BR		0.9977	3.48	0.0074	
				CMIS-GMDH		0.9989	2.53	0.006	
				CMIS-GP		0.9990	2.38	0.0057	

Table 3. Modeling and simulation of other gases' solubility in different ILs using AI methods

No.	Gas(es)	No. of IL	Inputs of the corresponding model	Type of model	Dataset	R ²	AARD%	RMSE	Ref.
				DBN (I)		0.9958	-	0.0118	
32	N ₂ O	25	T, P, Mw, T _c , P _c , ω	Cat-Boost (I)	Total	0.9959	-	0.0116	[125]
				ELM (I)		0.9931	-	0.0152	

				XGB (I)		0.9999	-	0.0016	
				DBN (II)		0.9964	-	0.011	
				Cat-Boost (II)		0.9954	-	0.0124	
				ELM (II)		0.9933	-	0.0149	
				XGB (II)		0.9998	-	0.0025	
33	N ₂ O	13	T, P, T _c , P _c , ω ,	CFNN-LM	Total	0.9994	-	0.0047	[126]
				RBFNN-PSO		0.9985	-	0.0073	
				GEP		0.9922	-	0.017	
34	N ₂ O	25	T, P, Mw, T _c , P _c	ANN	Test	0.9963	37.39	0.2980	[124]
				SVM		0.9970	16.40	0.1640	
				LSSVM		0.9959	29.80	0.3739	
35	SO ₂	8	T, P, Chemical structure of ILs	LSSVM	Total	0.9978	1.425	0.0103	[127]
36	SO ₂	8	T, P, T _c , P _c , ω	ANN	Test	0.9861	2.820	0.0217	[128]
37	CO ₂ , CH ₄	11	X _{CH₄} , X _{CO₂} , % CH ₄ (feed), %CO ₂ (feed), T, T _c , P _c , ω	CSA-LSSVM	Total	0.9846	3.041	0.3161	[129]
				PSO-ANFIS		0.9397	14.13	0.6253	
38	C ₂ H ₆	1	T, P, Mw _{gas} , ω _{gas}	ANN	-	-	21.86	-	[130]
	CH ₄					-	17.33	-	
	O ₂					-	15.80	-	
	N ₂					-	17.08	-	
	Ar					-	8.440	-	
	CO					-	16.74	-	

39	CO ₂ , H ₂ S,	11	T, P, mole fraction of IL, mole ratio	LSSVM	Total	0.9999	0.7545	0.0005	[131]
	SO ₂ , CH ₄ ,		of CO ₂ to second gas in feed, IL Mw,						
	H ₂ , N ₂ O		T _c , P _c , ω and second gas Mw, T _c , P _c						
40	CO ₂ , CH ₄ ,	40	T, P, Mw _{IL} , T _c , P _c , T _{c, IL} , P _{c, IL} , ω_{IL} ,	MLP-ANN	Total	0.9945	13.27	0.0173	[132]
	H ₂ S, N ₂ O,		Mw _{gas} , T _{c, gas} , P _{c, gas} , gas kinetic	Hybrid-ANFIS		0.9590	27.18	0.0447	
	SO ₂ , CO		diameter, ω_{gas}	PSO-ANFIS		0.9473	70.09	0.0510	
				CSA-LSSVM		0.9888	5.610	0.0245	

3. Results and discussion

3.1 Data acquisition

In this study, 3209 data points of gas absorption in various ILs were applied, and the main characteristics used to establish the models included the acentric factor (ω), M_w , P_c , pressure (P), temperature (T), and T_c . The parameters M_w , T_c , P_c , and ω were chosen because they reflect essential thermodynamic values that influence how gases dissolve in ILs. M_w , T_c , P_c , and ω are crucial for the ionic liquid's capacity to solvate gases. These characteristics are commonly utilized in thermodynamic models (such as PC-SAFT [133]) and in QSPR methods for forecasting gas solubility. Temperature and pressure serve as the main factors driving gas dissolution equilibrium. In addition, Numerous earlier studies utilized these parameters and demonstrated satisfactory outcomes [34, 100, 116, 125].

The full dataset, comprising 3209 experimental points, was partitioned into training (75%) and testing (25%) subsets via a random split. This method guarantees that the distributions of temperature, pressure, and IL types are proportionately maintained in both subsets. There was no grouping or selection conducted based on specific IL families, temperature ranges, or pressure intervals; rather, every sample had an equal chance of being allocated to either subset, which helps to prevent bias towards specific gas–IL combinations. The similarity in the R^2 , RMSE, and AARD values between the training and testing subsets indicates that the GP model captured underlying patterns rather than simply memorizing the training data. The details of data for CO_2 , CH_4 , N_2O and SO_2 absorption in different ILs are presented in Tables 4 and S1-S5 (**Supplementary Material**). In addition, the properties of different ILs and gases applied in the present study are shown in Table S6. The quality of modeling was assessed by statistical functions shown in Table S7; however, in this modeling study, average absolute relative deviation (%AARD), root mean squared error (RMSE), and R^2 were used as shown below:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i^{experimental} - y_i^{predicted})^2} \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n [y_i^{predicted} - y_i^{experimental}]^2}{\sum_{i=1}^n [y_i^{experimental} - y_m]^2}, \quad y_m = \frac{\sum_{i=1}^n y_i^{experimental}}{n} \quad (5)$$

$$\% AARD = \frac{100}{n} \sum_{i=1}^n \frac{|y_i^{predicted} - y_i^{experimental}|}{y_i^{experimental}} \quad (6)$$

Table 4. Experimental data of IL–CO₂ systems.

No.	Ionic liquid	Abbreviation	Temperature (K)	Pressure (bar)	Solubility (mole fraction)	No. Data	Ref.
1	1-ethyl-3-methylimidazolium acetate	[EMIM][Ac]	298.10-348.20	0.1000-19.996	0.09400-0.42800	27	[134]
2	1-butyl-3-methylimidazolium chloride	[BMIM][Cl]	353.15-373.15	25.540-369.46	0.13060-0.40600	45	[135]
3	1-ethyl-3-methylimidazolium dicyanamide	[EMIM][DCA]	303.20-343.20	1.4800-59.440	0.00862-0.21744	25	[136]
4	1-butyl-3-methylimidazolium thiocyanate	[BMIM][SCN]	292.35-384.15	10.500-315.00	0.12600-0.43000	56	[137]
5	1-ethyl-3-methylimidazolium tetrafluoroborate	[EMIM][BF ₄]	303.20-343.20	4.9600-43.290	0.04460-0.26560	25	[138]
6	1-butyl-3-methylimidazolium acetate	[BMIM][Ac]	283.10-348.20	0.1020-19.994	0.19200-0.45500	32	[139]
7	1-n-butyl-3-methylimidazolium nitrate	[BMIM][NO ₃]	313.15-333.15	15.470-93.170	0.16900-0.53000	21	[140]
8	1-butyl-3-methylimidazolium dicyanamide	[BMIM][DCA]	298.20-333.30	12.710-115.29	0.15430-0.56360	21	[141]
9	1-butyl-3-methylimidazolium tetrafluoroborate	[BMIM][BF ₄]	303.72-344.49	0.1773-0.8443	0.00236-0.01330	21	[142]
10	1-ethyl-3-methylimidazolium ethyl sulfate	[EMIM][EtSO ₄]	313.15-333.15	14.360-94.610	0.10000-0.45700	21	[140]
11	1-butyl-3-methylimidazolium methyl sulfate	[BMIM][MeSO ₄]	293.20-413.10	9.0800-98.050	0.03203-0.49150	54	[143]
12	1-hexyl-3-methylimidazolium tetrafluoroborate	[HMIM][BF ₄]	293.18-368.16	5.4000-866.00	0.10300-0.70300	104	[144]
13	1-ethyl-3-methylimidazolium hexafluorophosphate	[EMIM][PF ₆]	308.14-366.03	14.900-971.00	0.10400-0.61900	74	[145]
14	1-n-octyl-3-methylimidazolium tetrafluoroborate	[OMIM][BF ₄]	313.15-333.15	15.610-93.730	0.16000-0.70800	21	[140]
15	1-n-butyl-3-methylimidazolium hexafluorophosphate	[BMIM][PF ₆]	313.15-333.15	15.170-95.670	0.22800-0.72900	21	[140]
16	1-butyl-3-methylimidazolium trifluoromethanesulfonate	[BMIM][TfO]	298.20-333.30	10.390-114.77	0.10880-0.63630	27	[141]
17	1-hexyl-3-methylimidazolium hexafluorophosphate	[HMIM][PF ₆]	298.31-363.58	6.4000-946.00	0.09800-0.72700	98	[146]
18	1-octyl-3-methylimidazolium trifluoromethanesulfonate	[OMIM][TfO]	303.85-344.55	6.8000-340.00	0.21660-0.74140	65	[147]
19	1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[EMIM][Tf ₂ N]	312.10-543.15	6.2600-147.70	0.12300-0.59300	191	[148]
20	1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[BMIM][Tf ₂ N]	298.10-333.30	11.380-132.43	0.17590-0.75210	55	[141]
21	1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[HMIM][Tf ₂ N]	298.10-333.30	13.150-115.58	0.19930-0.75860	28	[141]
22	1-ethyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[EMIM][eFAP]	303.18-343.23	0.5362-0.7539	0.01730-0.03380	13	[149]
23	1-butyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[BMIM][eFAP]	303.16-343.27	0.6551-0.7691	0.02210-0.04010	15	[149]
24	1-hexyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[HMIM][eFAP]	298.13-343.08	0.5874-0.7330	0.02020-0.03980	12	[149]
Total						1072	

3.2. Model development

Koza [150] introduced an innovative evolutionary strategy termed GP for the automatic resolution of problems by characterizing the phenomena under scrutiny. Symbolic optimization was utilized to pinpoint potential solutions, employing a familiar tree-like representation [151]. This representation incorporated both terminals and functions, with functions representing mathematical algebraic operators and terminals signifying components of solutions and input parameters. The prospective solution, comprising a mix of functions and terminals, can be represented through an expression tree. Various genetic operators, including mutation, crossover, and selection, are regularly employed to broaden the pool of potential solutions. This iterative process culminates when the system attains a predefined number of iterations or unearths a satisfactory solution. In this context, a fitness function was established to assess solution quality, primarily by utilizing the MSE, which is targeted for minimization[152].

While the mathematical intricacies of the correlations generated by GP are complicated, they present several advantages that black-box models fail to deliver. To begin with, the explicit equation enables users to calculate gas solubility directly without the need for training datasets, specialized machine learning software, or model retraining, thus making the correlation easily portable for use in industrial applications or offline scenarios (such as spreadsheets, process simulators, and numerical codes). Additionally, the functional format offers full transparency, allowing users to examine how temperature, pressure, and IL characteristics influence the model, which is not achievable with MLP-ANN or LSSVM frameworks. Moreover, the GP equation can be integrated into design calculations or utilized within process optimization frameworks as a closed-form function, a capability that is largely not possible with most black-box methodologies. Consequently, despite its mathematical complexity, the GP correlation provides greater interpretability, reproducibility, and practical applicability compared to traditional black-box models.

In this study, an open-source programming toolbox named GPTIPS, prepared by Searson to perform multi-gene symbolic regression in the MATLAB environment [153], was applied to develop the non-linear correlation. The significant parameters of the GP model, which are obtained by trial and error, are shown in Table 5. In this, the maximum depth of trees was 8 for all gases except 10 for CO₂. The absolute minimum and maximum numbers of genes allowed in an individual were 2 and 8. The minimum and maximum numbers of generations to run were 2000 and 2500, with population sizes of 300 and 500, respectively. The functions used were times, minus, plus, divide, square, exp, sin, cos, tan and cot.

To improve the GP model and remove arbitrary hyperparameter selections, a systematic multi-stage tuning strategy was utilized. First, wide parameter ranges were defined based on values from previous symbolic regression studies (population size 50-600, tree depth 6-12, genes 2-10, generations 1500–3000). The GP was then sequentially trained using various configurations within these specified ranges. Each configuration was evaluated based on (i) the consistency of R^2 , RMSE, and AARD% between training and testing, (ii) the stability of convergence across multiple runs with different random seeds, and (iii) the simplicity of the model to avoid unnecessary equation complexity. Hyperparameters were selected only when they provided stable convergence and similar accuracy across repeated trials, thus guaranteeing reproducibility while steering clear of both underfitting and overfitting.

A simple-to-use Excel file was developed to calculate the solubility using the input parameters. The users can put the data in each cell and calculate the solubility easily. The developed GP equations are as follows:

N₂O Solubility

$$\begin{aligned}
 &= \frac{2185.0 \left(2P + \sin \omega + \frac{M_\omega}{T} \right)}{T - 2\omega + \cot(3M_\omega) + \cot(\sin(\sin M_\omega) - \cos(P_c) - 16.68)(T + 3P + 2\omega + \cos((M_\omega + \cos(P_c) + 3.419)(M_\omega - T + \omega))) + \sin(\cot(T + \omega)) + \cos(\cot(\omega)) +} \\
 &\quad + \tan(\cot(2M_\omega + \omega)) + \sin(\cot(\sin P)) + \cos(T_c) + 2 \cos(P_c) \\
 &\quad - \frac{6167P}{T + \omega + \sin(\cot(T + \cot(\omega) - 7.329) + \sin(\cot(T - 7.394)) + \sin(T_c + \omega) + \cos(P_c)(T + 5P +} \\
 &\quad + M_\omega - P_c + \cot(\omega^2) + \sin(P + P_c + \cot(\omega)) + \sin(\cot(2P)) + \cot \omega + \sin\left(\cot\left(\frac{T - 7.332}{\sin P}\right)\right) - \frac{T_c}{T} - 23.7) - 0.04292
 \end{aligned} \tag{8}$$

CH₄ Solubility

$$\begin{aligned}
 &= (1.457 \times 10^{-8})(-T_c - P_c - 38.36\omega - 9.566)^2 - (1.457 \times 10^{-8})(-M_\omega - 5.167P_c)^2 - (1.457 \times 10^{-8})(2.793997 - 2P_c)^2(0.181P + M_\omega - 38.36\omega + \tan(M_\omega)) \\
 &\quad - (1.457 \times 10^{-8})T(T_c - P_c^2) \\
 &\quad + (1.457 \times 10^{-8})P(P_c - M_\omega)^2 \\
 &\quad - \frac{(1.082 \times 10^{-13})(M_\omega + 12.11M_\omega^2 + T_c^2 + PM_\omega^2 + 5.167)^2(5.167T + e^{\cot \omega} + \tan(7.00883 - M_\omega^2) + \tan(7.008659 - (10.001158M_w)^2 - (6.239656)^2)}{(\omega - T_c - P + 2(10.370932)^2)^2 + \frac{T_c^2 + 6.24T(P - 10.34)}{(7.093678 - P_c)^2 - 6.941}} \\
 &\quad + \frac{(1.021 \times 10^{-7})(M_\omega - P + P_c)^2(P - 9.962)(T_c - P + P_c + (-5.524P_c)^2)}{P(6.840296 - P_c)^2(7.009T - T_c)(0.181P + 2P_c + \tan(M_\omega) - 7.001)} + 1.747 \times 10^{-15}
 \end{aligned} \tag{9}$$

H₂S Solubility

$$\begin{aligned}
 &= \frac{(3.858 \times 10^5)(7P + 2.63)}{T^2(T + 12P + P_c + \cot(\tan P_c) + \cot T - 42.93)} \\
 &\quad - \frac{9754(3P + \cos(P_c + \omega^2) + 4.198)}{(T + 11P + \tan(\tan M_\omega) + 3 \tan(\tan(P_c)))(21.72P + 7.241M_\omega + 7.241 \tan(\tan(M_\omega))) + 14.48 \sin(T) + 7.241 \sin(P) + 7.241 \tan(M_\omega) + 261.3)} \\
 &\quad + 0.0008746
 \end{aligned} \tag{10}$$

CO₂ solubility

$$\begin{aligned}
&= 0.2819 \left(e^{-\frac{P_c + \frac{6.939 e^{\frac{6.991}{P}} + \frac{\tan(T_c + 0.02439)}{P_c} \omega}{\frac{\tan(-P_c - 7.016)}{P_c}}}{P}} \right)^2 - \frac{3.47}{\frac{\tan(T_c - P_c)}{e^{\frac{6.939}{P_c} \frac{P}{T}} + \frac{e^e}{e^e} \left(\frac{\tan(\tan(T_c))}{P_c} + 5.836 \right)}} - \frac{0.5395}{\frac{\tan(-M_\omega - 5.939)}{7.016} M_\omega + \frac{\tan(\tan(M_\omega - P))}{P_c}} - \frac{0.8051}{\frac{\tan(-P_c - 6.939)}{+T_c + \frac{0.1769 T_c}{5.654}}}{e^{P_c + \frac{e^{\frac{5.654}{P}}}{2P_c - T - \tan(\cot(T))}}} \\
&+ 1.25 e^{\frac{\tan(T_c - \omega)}{T - \tan(-M_\omega - 6.939)}} + \frac{1.25}{\frac{\tan(-P_c - 6.939)}{P_c + \frac{\tan(-P_c - 6.939)}{\tan(\frac{P}{M_\omega} - T_c^2) + 6.059}}}{e} - \frac{0.1796}{\frac{\tan(-P_c - 5.72)}{e^{P_c - e^{\tan(T_c + 7.016) - \left(\tan(\frac{\sin(P_c)}{P}) \right)^2}}}} - \frac{0.01368 \sin\left(\frac{P_c}{T} - 0.02439 M_\omega e^{e^{-\frac{5.654}{P}}} + 5.836\right)}{P} \\
&+ \frac{0.5357 M_\omega}{e^{P e^{\frac{6.939}{P_c} \frac{\omega}{T}} \left(\frac{\cot(M_\omega)}{P} \right)^2 + 6.931}} + \frac{7072}{T^2} \left(e^{-\frac{P_c}{T + 41 \tan(T_c - \omega)}} + e^{\frac{\tan(-P_c - 6.939)}{M_\omega}} \right) + \frac{13333 M_\omega e^{e^{-\frac{5.619}{P}}}}{T^3 e^{\frac{5.654}{P}}} + 2.365
\end{aligned}
\tag{11}$$

SO₂ Solubility

$$\begin{aligned}
&= 0.15P + 0.15 \sin \left(\sin \left(\sin \left(P + \frac{T - M_\omega}{M_\omega + 9.446666} \right) \right) \right) \\
&+ \frac{8284.0}{(T + \cot(T) - 10.24)(T + \cot(T) - 10.22) - (\sin(P))^2(10.01T_c + M_\omega P_c) + \frac{T + \cot(\cot(T)) - \frac{10.22}{P_2} - 10.01}{\sin(M_\omega - 9.219) + \frac{T - M_\omega}{M_\omega}}}
\end{aligned}$$

$$\begin{aligned}
& + \frac{0.15 \left(T + 21.86M_\omega + P_c - \cot\left(\frac{P}{\omega}\right) - \cot(M_\omega + 5.888006) \right)}{M_\omega + \sin(\cot(T)) + \sin(TM_\omega) + \sin(PM_\omega) + \sin(P_c)} + \frac{14644.0 \left(\frac{3P}{P_c} - \omega + 5.356 \right)}{(T - 30.65)(T - 30.47) + \frac{2T + M_\omega + P_c}{P} + \frac{M_\omega + \cot\left(\frac{P}{P_c}\right) - 18.86}{\omega + \sin(M_\omega + \omega)}} - 4.264 \\
\end{aligned} \tag{12}$$

CO solubility

$$\begin{aligned}
& = (9.138 \times 10^{-9})P(T + T_c) - (4.569 \times 10^{-9})TM_\omega + (4.569 \times 10^{-9})T_cP_c + (1.631 \times 10^{-8})\tan(PM_\omega)(3P + T_c + 3(\tan(T))^2) + (4.569 \times 10^{-9})PM_\omega^2 \\
& - (4.569 \times 10^{-9})P^2M_\omega + (1.631 \times 10^{-8})P(2T + P + T_c + (\tan(T))^2 + \cot(TP) + \cot(M_\omega + T_c - TP)\cot(M_\omega(T + T_c))) + \frac{(1.631 \times 10^{-8})T \tan(M_\omega(T_c + P_c))}{\tan(PT_c)} \\
& + \frac{(4.569 \times 10^9)(T + 8.363774)(T + T_c)}{P} - \frac{(4.569 \times 10^9)(T + 3T_c + M_\omega^2)(T_c + P_c - 3PM_\omega)}{M_\omega \tan(P_c + \omega)} + \frac{(4.569 \times 10^{-9})(\tan(T_c))^2(\cot(TP) - TP)}{M_\omega \tan(\sin(P_c - 8.577))} \\
& + \frac{(1.631 \times 10^{-8})M_\omega^2(\tan(T_c) - PM_\omega)}{M_\omega \tan(P_c + \omega)} - 0.002124
\end{aligned} \tag{13}$$

Table 5. GP setting parameters employed in the current research.

Parameter	Value (CO ₂)	Value (other gas)
Population size	500	300
Number of generations	2500	200
Maximum depth of trees	10	8
The maximum number of genes in an individual	10	2
Operators used	+, -, ×, /, exp, .^2 square, sin, cos, tan, cot	+, -, ×, /, exp, .^2 square, sin, cos, tan, cot

3.3. Model accuracy and validation

According to Figure 1, congestion of data near the Y=X line confirms the considerable agreement between the actual values and those estimated by the two techniques. Moreover, the R^2 values for the GP model prediction of CO₂, CH₄, CO, H₂S, N₂O and SO₂ solubilities in ILs are 0.9795, 0.9971, 0.9997, 0.9894, 0.9950 and 0.9774, respectively, suggesting a statistically very strong fit between experimental and predicted values. A thorough analysis of Figure S3 demonstrated accurate monitoring of the trend of target values by the GP models. As shown in Figure S4, the maximum values of relative deviation for the prediction of CO₂, CH₄, CO, H₂S, N₂O and SO₂ solubilities in ILs by the GP model were 3.2, 0.97, 1.47, 1.05, 0.9 and 1.03, respectively, while the minimum values were -10.65, -2.076, -0.39, -1.5, -13.15 and -4.3, respectively. Higher deviation was observed at lower solubilities. Figures S5-S10 show the histogram of errors for the prediction of CO₂, CH₄, CO, H₂S, N₂O and SO₂ solubilities in ILs by the GP model. As shown in Table 6, the R^2 values for all cases were more than 0.9. In addition, the %AARD values for all predictions were less than 20, demonstrating that the model's predictions are within 20% of the experimental values.

While the final GP correlations may appear mathematically intricate, this intricacy is indicative of the significant nonlinearity inherent in gas-IL systems rather than a result of overfitting. The model was

developed using a substantial and varied dataset, and the strong alignment of R^2 , RMSE, and AARD% between the training and testing subsets suggests that the produced expressions generalize effectively without excessive variance. GP enhances the model structure only when additional terms lead to a discernible decrease in prediction error, indicating that the complexity is not arbitrary but stems from the physical and thermodynamic behaviors captured in the data. Importantly, even though the GP equations are lengthy, they provide a closed-form correlation that can be readily utilized in spreadsheets, engineering simulators, or design tools without the need for ML software, stored weight matrices, or retraining. Therefore, the complexity of the final expression is warranted by its precision, portability, and clarity, offering benefits that conventional black-box models do not provide.

Although GP generates explicit mathematical formulas, the resulting expressions tend to be complex due to the need to account for the highly nonlinear relationship between gas solubility and factors like temperature, pressure, and the critical properties of ionic liquids. Consequently, the symbolic relationships produced by GP do not translate easily into straightforward, interpretable physicochemical principles. Instead, their significance lies in offering clear, equation-based predictions while still demonstrating physically meaningful behavior. This is illustrated in Figure 2, where the GP model accurately captures the essential thermodynamic patterns. As shown in Figure 2, the gas solubility in ILs was increased by increasing pressure and decreasing temperature. However, there is an exception for CO solubility in [BMIM][MeSO₄] [154], which was increased by increasing temperature, which has been mentioned as a rare phenomenon [154]. Thus, although the GP equations do not provide new mechanistic insights into the interactions between gas and ionic liquids, they effectively encapsulate and reflect the anticipated physical trends across a diverse range of systems with great precision.

The differences in predictive performance across gases can be attributed to the inherent thermodynamic behavior and statistical properties of each dataset, such as the number of data points. For CO, the solubility patterns demonstrate consistent trends and almost linear monotonic trends concerning

temperature and pressure for most ionic liquids (ILs), with the experimental value range being quite narrow and in low amounts, as can be seen from Table S5. This regularity in the CO dataset enables the GP model to capture its characteristics with near-perfect precision. In contrast, N₂O exhibits more intricate data and as Table S3 indicates that N₂O mole fractions range from 0.001-0.781, covering almost three orders of magnitude, with large variability across ionic liquids, temperature, and pressure. The N₂O dataset also features regions of very low solubility, where even minor absolute variations can lead to substantial percentage errors, thus elevating the AARD despite having reasonable RMSE and R² values. These factors together account for the GP model's exceptional performance with CO while demonstrating comparatively higher errors for N₂O.

Table S7 demonstrates the comparison of the statistical parameters of this study and the previous study [132], which used a CSA-LSSVM model to predict gas solubility in SO₂, CH₄, N₂O, CO, H₂S and CO₂. The prediction accuracy of CSA-LSSVM is higher in all gases except SO₂, in which GP showed higher accuracy. However, the GP model is easier to use than complex AI models.

The heat map displayed in Fig. 3 illustrates the AARD% values for each Gas–IL system, facilitating a visual assessment of the prediction challenges associated with different gases. Darker shades indicate greater deviations, while lighter shades signify more precise predictions. The heat map distinctly indicates that gases like CO, H₂S and SO₂ generally show the least deviations across most ILs. Conversely, CO₂ reveals higher AARD% values for various ILs, indicative of their wider solubility ranges.

To tackle the issue of model generalization beyond a straightforward train–test division, the ability of the GP model to extrapolate was assessed using independent experimental data sourced from existing literature that was not utilized during any phase of model creation and the training data. The findings regarding extrapolation are presented in Table S8. The GP model showed acceptable predictive capability

on these previously unseen datasets, showing acceptable statistical accuracy. This indicates that the model captures the fundamental physicochemical trends that influence gas solubility in ILs, rather than simply interpolating the training data. It should be noted that in extrapolation, an outlier detection procedure was applied to the external datasets to ensure robustness. Data points identified as inconsistent or erroneous in the original literature were excluded in accordance with established practices in machine learning-based thermophysical modeling. The complete extrapolation datasets are available in the Supplementary Excel file to ensure transparency. It is important to mention that we could only find two independent experimental data points related to CO available in the literature. Due to this very limited dataset, conducting a statistically valid validation of the model was not possible, and traditional performance metrics (such as R^2) cannot be considered reliable in these circumstances. As a result, the typically low R^2 value noted for CO does not indicate inadequate model performance but instead highlights the lack of sufficient experimental data. This constraint is clearly recognized in the manuscript.

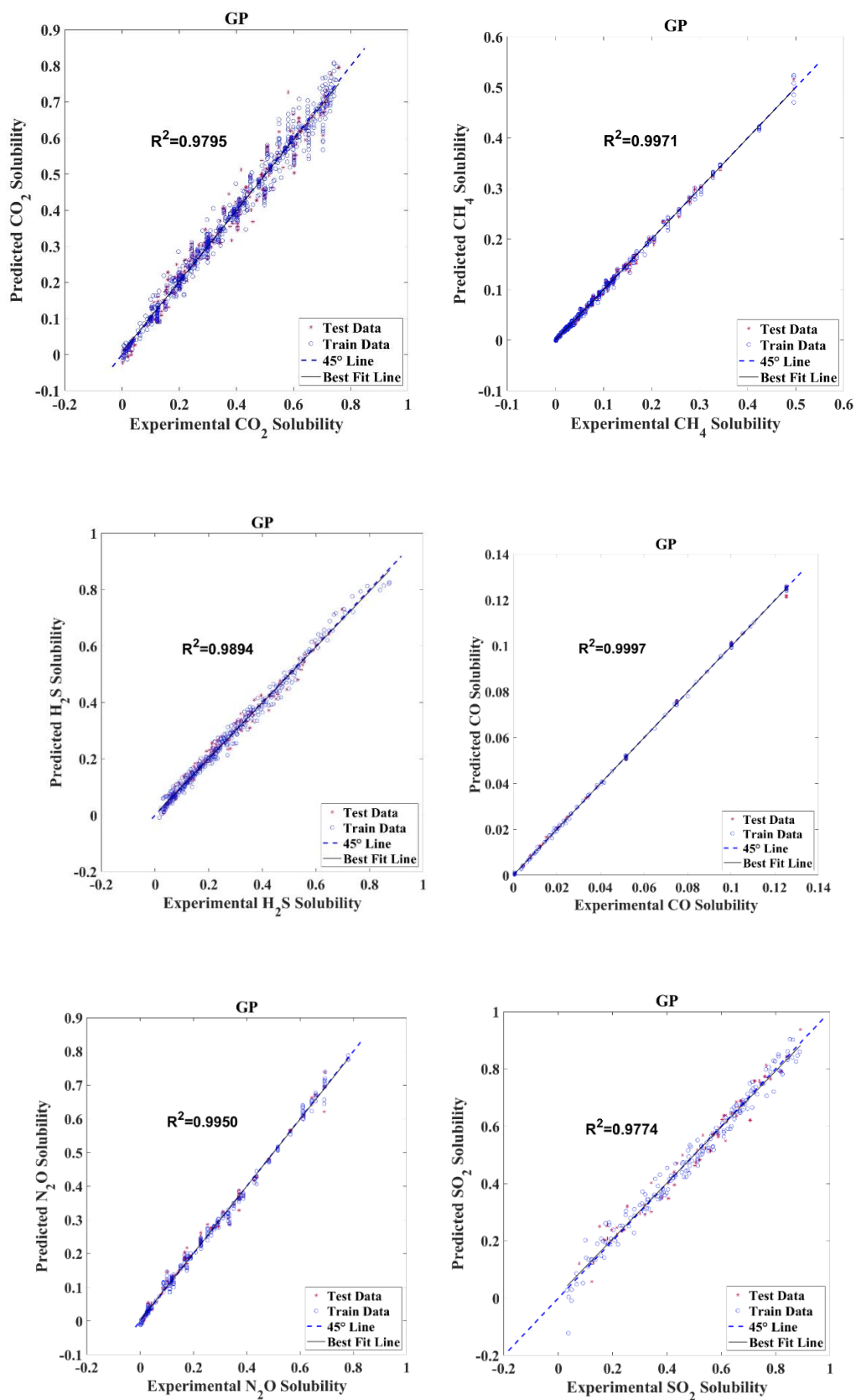
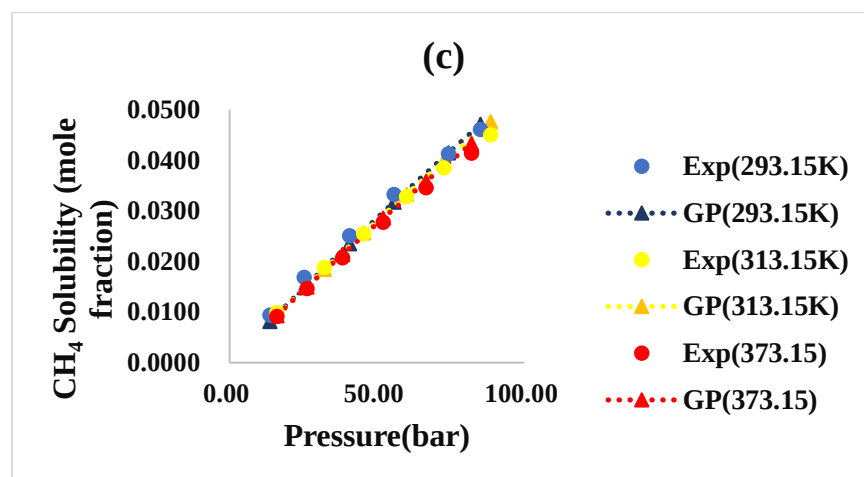
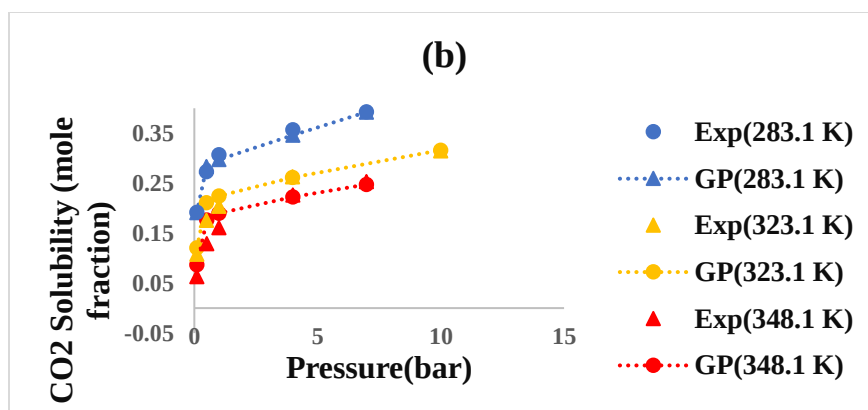
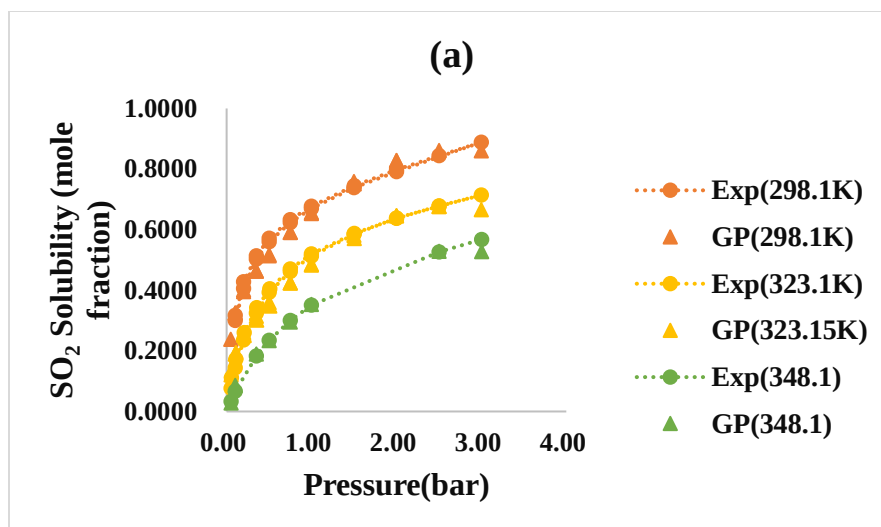


Fig. 1. Experimental values of different gas solubility in ILs versus predicted data.



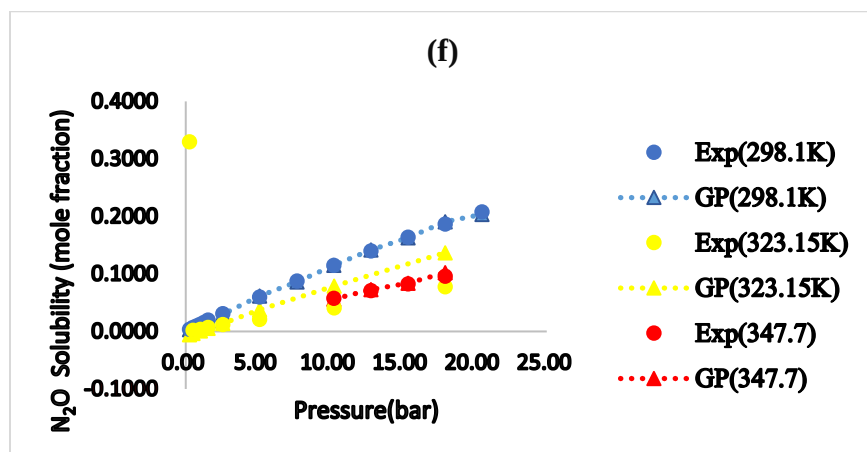
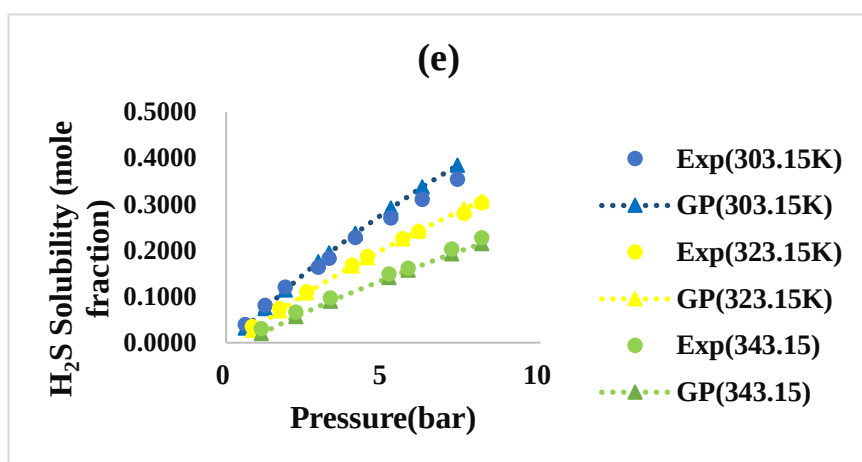
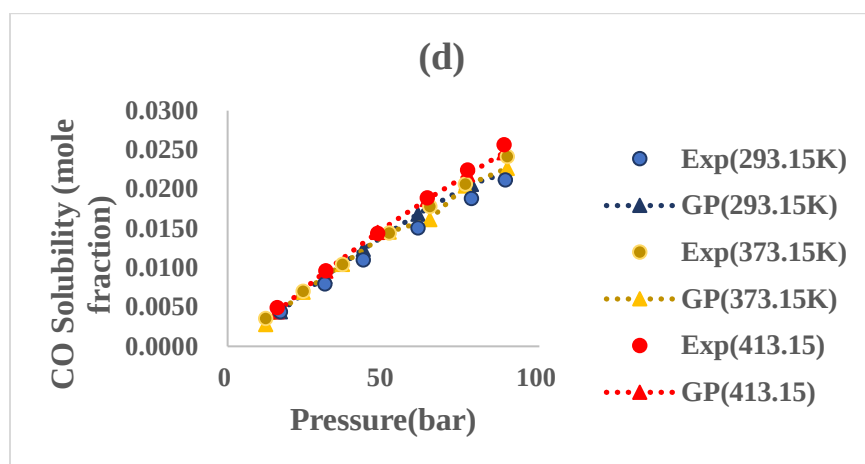


Fig. 2. Effects of temperature and pressure on gas solubility in ILs; a) SO₂ absorption in [BMIM][MeSO₄][155], b) CO₂ absorption in [BMIM][AC][139], c) CH₄ absorption in [BMIM][MeSO₄][156], d) CO absorption in [BMIM][MeSO₄][154], e) H₂S absorption in [BMIM][BF₄][157] and f) N₂O absorption in [BMIM][BF₄][158].

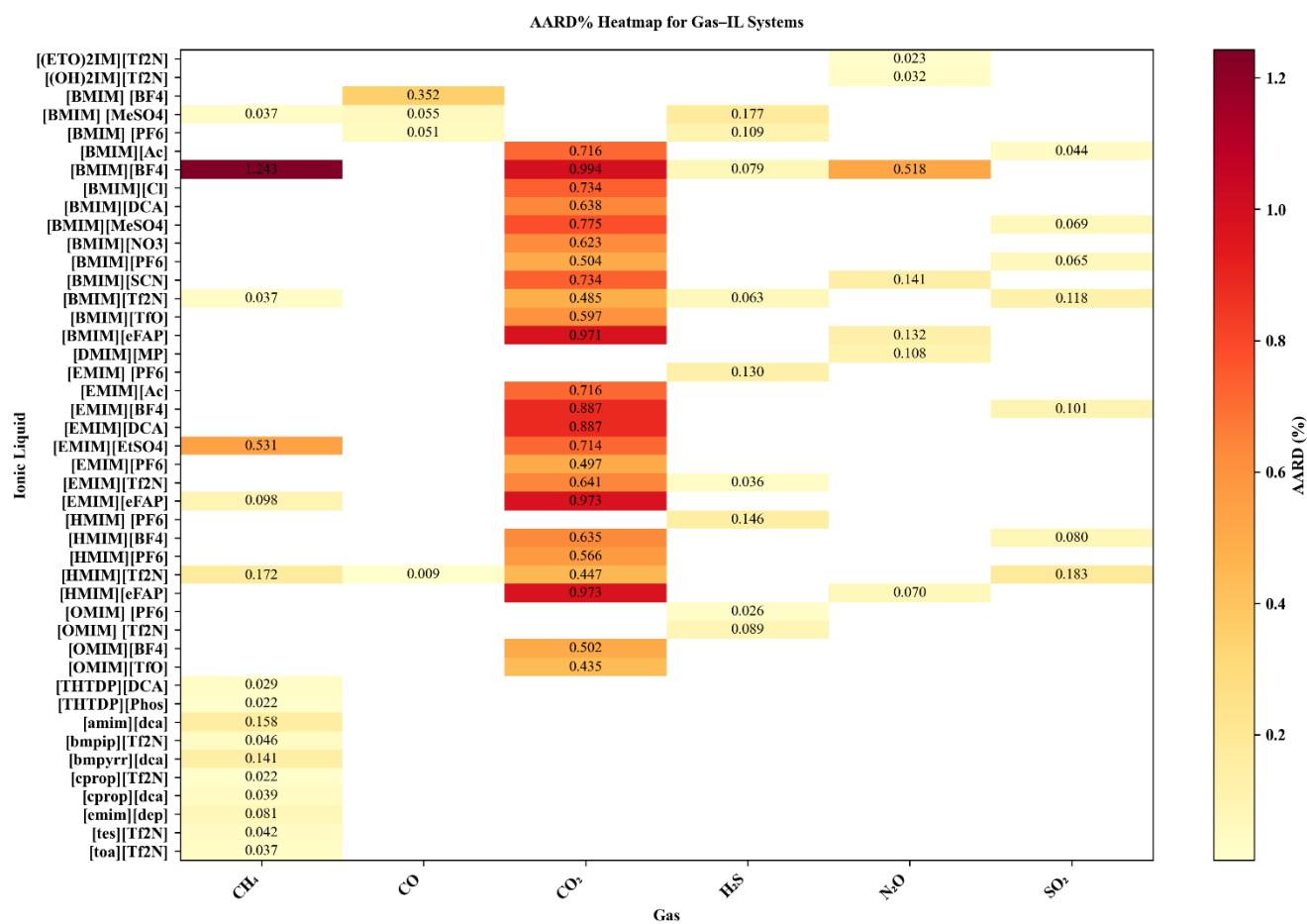


Fig. 3 AARD% heatmap for Gas-IL systems

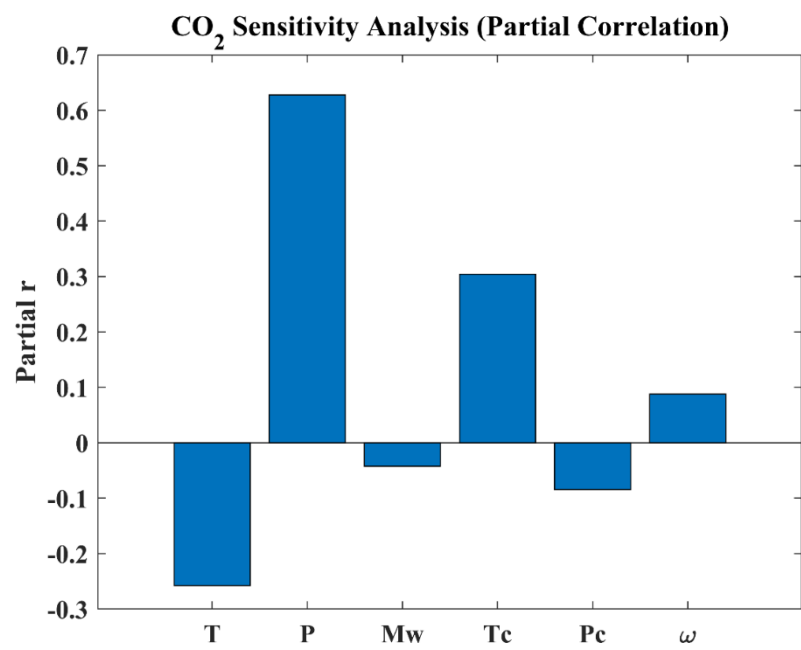
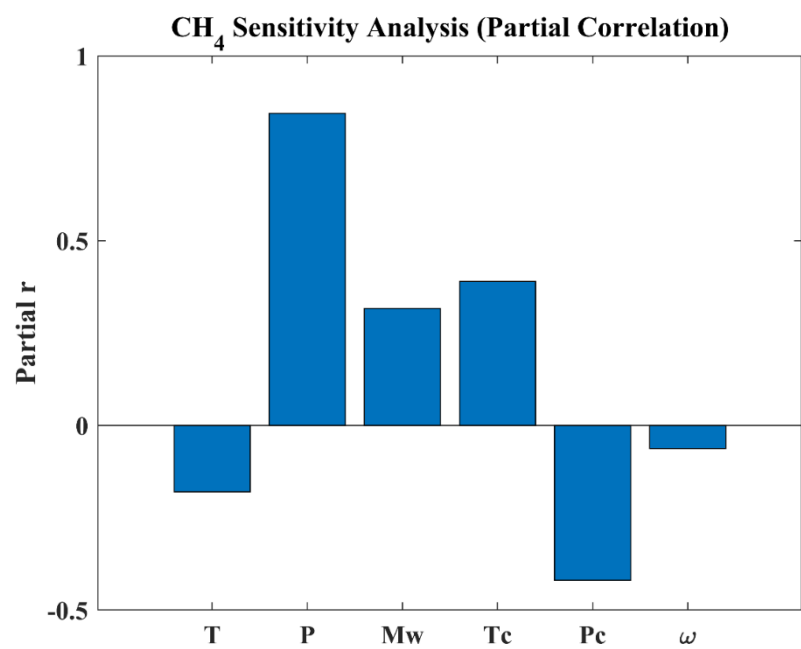
Table 6. Statistical parameters of the developed model.

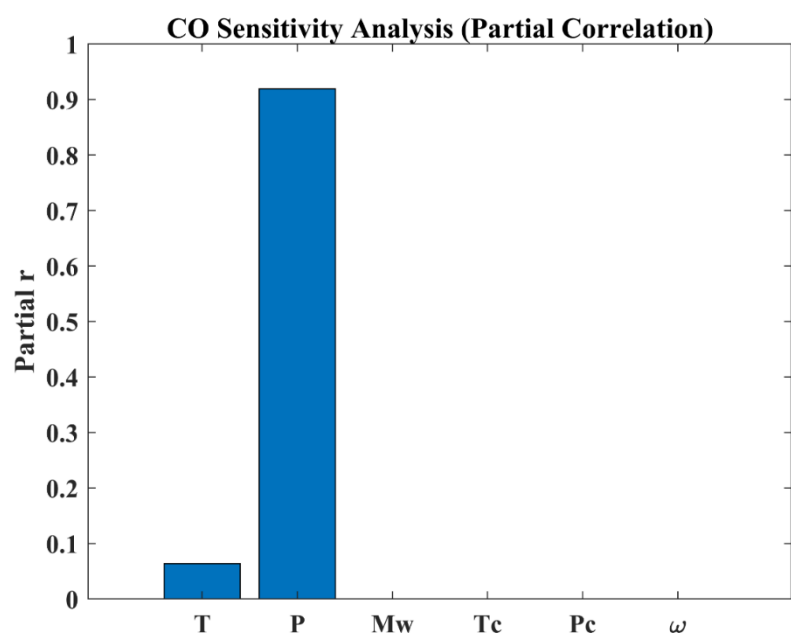
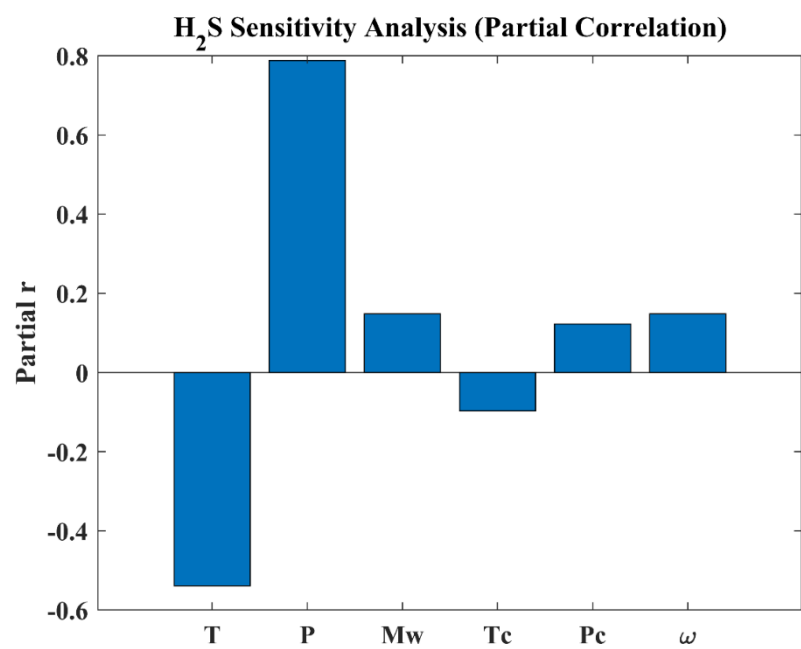
Parameter	Dataset					
	Train	Test	Total	Train	Test	Total
	CO ₂			H ₂ S		
R ²	0.9806	0.9758	0.9795	0.9904	0.9856	0.9894
RMSE	0.0284	0.0307	0.0290	0.0181	0.0196	0.0185
AARD%	10.7162	15.6190	11.9419	9.1548	8.8881	9.0884
	CO			CH ₄		
	Train	Test	Total	Train	Test	Total
	CO			CH ₄		
R ²	0.9998	0.9994	0.9997	0.9974	0.9965	0.9971
RMSE	0.0006	0.0011	0.0008	0.0051	0.0057	0.0052
AARD%	4.7568	4.3511	4.6548	10.5031	9.7038	10.3030
	N ₂ O			SO ₂		
	Train	Test	Total	Train	Test	Total
	N ₂ O			SO ₂		
R ²	0.9958	0.9927	0.9950	0.9793	0.9716	0.9774
RMSE	0.0135	0.0172	0.0145	0.0326	0.0359	0.0335
AARD%	19.1969	10.6916	17.0650	10.9012	9.1633	10.4704

3.4. Feature importance

We carried out a sensitivity analysis to assess the impact of each input parameter on gas adsorption in ILs. To this end, we exploited partial correlation coefficients (r) and evaluated the effect of various input parameters on gas solubility. In the end, large r values for an input parameter demonstrated considerably greater effect of that parameter on the relevant output, which was gas solubility. The feature importance of the input parameters was determined by applying the partial correlation coefficients method [159, 160].

Figure 4 demonstrates the effects of various input parameters on the gas's solubility in ILs. In all cases, increasing pressure has the utmost effect in increasing the solubility. Temperature has a negative effect on the gas solubility in ILs except for CO. This is consistent with previous studies reporting that CO in certain ionic liquids (e.g., [BMIM][CH₃SO₄]) can display a weak or even slightly positive solubility-temperature dependence, a rare but documented phenomenon [154]. Also, the data for CO solubility in ILs is rarer than other gases. Other parameters demonstrate little sensitivity (except T_c for CH₄ and P_c for CO₂) as they exhibit only slight variations within each group of gases and are strongly correlated with each other, leading to restricted independent explanatory capability in the context of partial correlation analysis.





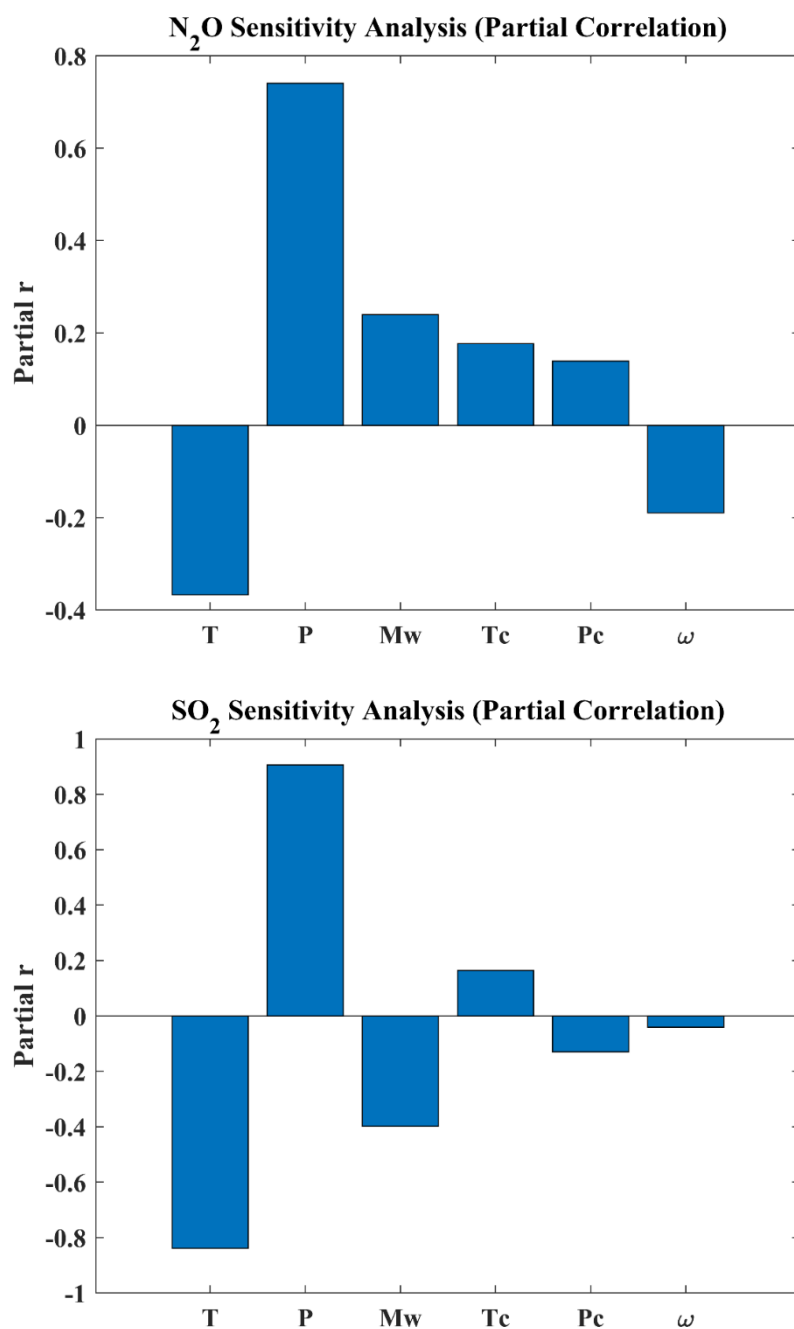


Fig. 4. Sensitivity analysis of the studied systems

Conclusions

This research reviewed the recent progress in using ML models for the estimation of pollutant gas absorption in ILs. A GP model was proposed to predict the solubility of SO₂, CH₄, N₂O, CO, H₂S and CO₂ in ILs, by considering the parameters of ω , T_c, P_c, and M_w of ILs as well as P and T of the process.

The calculations agreed well with the experimental data with high accuracy. In addition, the GP model proposed can precisely calculate the sour gas solubility in ILs with low dependent parameters, so that the conditions and behavior of the system can be monitored. Furthermore, a simple-to-use Excel method was created that can easily estimate the solubility of polluting gases in different ILs with high accuracy. The novel model provides a precise method of calculating gas solubility in ILs, which can be utilized by scientists and engineers as a tool in the accurate prediction of highly polluting gases in ILs under different operating conditions. The tool developed will be highly valuable in supporting research and application of ILs in combating global warming and air pollution.

Declaration of generative AI in scientific writing

During the preparation of this work, the author(s) used Grammarly and ChatGPT in order to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Abbreviations

Full name	Abbreviation	Full name	Abbreviation
Average absolute relative deviation	AARD	1,3-diethoxyimidazolium bis(trifluoromethylsulfonyl)imide	[(ETO)2IM][Tf2N]
Adjusted coefficient of determination	Adj. R_D^2	1,2,3tris(diethylamino)cyclopropenylum bis(trifluoromethylsulfonyl)imide	[CPROP][Tf2N]
Adjusted Pearson coefficient of determination	Adj. R^2	1,2,3tris(diethylamino)cyclopropenylum dicyanamide	[CPROP][DCA]
Absolute error	AE	1,3-diethoxyimidazolium bis(trifluoromethylsulfonyl)imide	[(ETO)2IM][Tf2N]
Artificial intelligence	AI	1,3-dihydroxyimidazolium bis(trifluoromethylsulfonyl)imide	[(OH)2IM][Tf2N]
Absolute mean square relative error	AMSRE	1,3-dihydroxyimidazolium bis(trifluoromethylsulfonyl)imide	[(OH)2IM][Tf2N]
Adaptive neuro fuzzy inference system	ANFIS	1,3-dimethylimidazolium methyl phosphonate	[DMIM][MP]
Artificial neural networks	ANN	1,3-dimethylimidazolium methyl phosphonate	[DMIM][MP]
Absolute percentage error	APE	1-allyl-3-methylimidazolium dicyanamide	[AMIM][DCA]
Bayesian Regularization	BR	1-butyl-1-methylpiperidinium bis(trifluoromethylsulfonyl)imide	[BMPIP][Tf2N]
Classification and Regression Tree	CART	1-butyl-1-methylpyrrolidinium dicyanamide	[BMPYRR][DCA]
Categorical boosting	Cat-Boost	1-butyl-3-methylimidazolium acetate	[BMIM][Ac]
Cascade Fee-Forward Neural Network	CFFNN	1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[BMIM][Tf2N]
Cascaded Forward Neural Network	CFNN	1-butyl-3-methylimidazolium chloride	[BMIM][Cl]

Committee Machine Intelligent System	CMIS	1-butyl-3-methylimidazolium dicyanamide	[BMIM][DCA]
Convolutional Neural Network	CNN	1-butyl-3-methylimidazolium methyl sulfate	[BMIM][MeSO4]
Conductor-like Screening Model	COSMO	1-butyl-3-methylimidazolium tetrafluoroborate	[BMIM][BF4]
Cubic-Plus Association	CPA	1-butyl-3-methylimidazolium thiocyanate	[BMIM][SCN]
critical point-based perturbed chain theory	CP-PC-SAFT	1-butyl-3-methylimidazolium trifluoromethanesulfonate	[BMIM][TfO]
Coupled simulated annealing	CSA	1-butyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[BMIM][eFAP]
deep belief network	DBN	1-ethyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[EMIM][eFAP]
Deep belief network	DBN	1-ethyl-3-methylimidazolium acetate	[EMIM][Ac]
Deep neural network initialization with decision trees	DJINN	1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[EMIM][Tf2N]
Deep Neural Network	DNN	1-ethyl-3-methylimidazolium dicyanamide	[EMIM][DCA]
Error	E	1-ethyl-3-methylimidazolium diethyl phosphate	[EMIM][DEP]
Extreme Learning Machine	ELM	1-ethyl-3-methylimidazolium ethyl sulfate	[EMIM][EtSO4]
equation of state	EoS	1-ethyl-3-methylimidazolium hexafluorophosphate	[EMIM][PF6]
electrolyte perturbed-chain SAFT	ePC-SAFT	1-ethyl-3-methylimidazolium tetrafluoroborate	[EMIM][BF4]
Genetic Algorithm	GA	1-ethyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[EMIM][eFAP]

gradient boosting trees	GBDT	1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[HMIM][Tf2N]
Gradient boosting machine	GBM	1-hexyl-3-methylimidazolium hexafluorophosphate	[HMIM][PF6]
group contribution	GC	1-hexyl-3-methylimidazolium tetrafluoroborate	[HMIM][BF4]
Gene expression programming	GEP	1-hexyl-3-methylimidazolium tris(pentafluoroethyl)trifluorophosphate	[HMIM][eFAP]
greenhouse gases	GHGs	1-n-butyl-3-methylimidazolium hexafluorophosphate	[BMIM][PF6]
Group Method of Data Handling	GMDH	1-n-butyl-3-methylimidazolium hexafluorophosphate	[BMIM][PF6]
genetic programming	GP	1-n-butyl-3-methylimidazolium nitrate	[BMIM][NO3]
Genetic programming	GP	1-n-octyl-3-methylimidazolium tetrafluoroborate	[OMIM][BF4]
Generalized regression neural network	GRNN	1-octyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[OMIM][Tf2N]
pentafluoroethane	HFC-125	1-octyl-3-methylimidazolium hexafluorophosphate	[OMIM][PF6]
1,1,1,2-tetrafluoroethane	HFC-134a	1-octyl-3-methylimidazolium trifluoromethanesulfonate	[OMIM][TfO]
difluoromethane	HFC-32	acentric factor	ω
Imperialist competitive algorithm	ICA	Argon	Ar
ionic liquids	ILs	Average bonding information content	$\overline{BIC}$
Ionic Solubility Compound Model	ISCM	carbon dioxide	CO ₂
Kling Gupta's efficiency	KGE	Carbon monoxide	CO
Levenberg-Marquardt	LM	coefficient of determination	R ²
Linear Regression	LR	critical pressure	P _c
Least square Support Vector Machine	LSSVM	critical temperature	T _c

Long Short-Term Memory	LSTM	electrostatic potential surface	SEP
Mean absolute deviation	MAD	Hydrogen sulfide	H ₂ S
Mean absolute error	MAE	Krypton	Kr
Absolute average deviation	MAPE	methane	CH ₄
Mean absolute percentage error of the average data	MAPEA	methyltriocetylammmonium bis(trifluoromethylsulfonyl)imide	[TOA][Tf ₂ N]
Mean error	ME	Nitrous oxide	N ₂ O
machine learning	ML	sulfur dioxide	SO ₂
Multilayer Perceptron	MLP	Triethylsulfonium bis(trifluoromethylsulfonyl)imide	[TES][Tf ₂ N]
Multilinear Regression	MLR	trihexyltetradecylphosphonium bis(2,4,4- trimethylpentyl) phosphinate	[THTDP][Phos]
Mean relative percentage deviation.	MRPD	Trihexyltetradecylphosphonium dicyanamide	[THTDP][DCA]
Mean Squared Error	MSE	Xenon	Xe
Mean square error	MSE		
Normalized mean squared error	NMSE		
Non-random two-liquid	NRTL		
number of single bonds	Nsb		
Perturbed-Hard-Sphere Theory	PHST		
perturbed hard trimmer chain	PHTC		
Peng-Robinson	PR		
Percentage of root mean square relative error	PRMSRE		
Particle swarm optimization	PSO		
Quantitative-structural property relationship	QSPR		
Pearson correlation coefficient	R		
Square of Pearson correlation coefficient	R ²		

Relative absolute error	RAE
Resilient Backpropagation	RB
Radial Basis Function Neural Network	RBFNN
restricted Boltzmann machines	RBM
Regression correlation coefficient	RC
Linear correlation coefficient	RD
Root mean square error	RMSE
Recurrent Neural Network	RNN
Relative percentage error	RPE
Root relative squared error	RRSE
room-temperature ionic liquid	RTIL
Sum of absolute error	SAE
Statistical Associating Fluid Theory	SAFT
Sum of absolute percentage error	SAPE
Shuffled Complex Evolution	SCE
Sum of error	SE
Square error	SE
Standard error	SE
Standard error of prediction	SEP
Soave-Redlich-Kwong	SRK
Sum of relative percentage error	SRPE
Sum of squares error	SSE
Standard deviation of error	STD
Support vector machine	SVM
Statistical Wertheim-Tanh Chain Fluid	fSWCF
temperature-dependent Perturbed-Chain	tPC-PSAFT
Statistical Associating Fluid Theory	
Universal Quasi-chemical Functional- Group Activity Coefficients	UNIFAC

vapor-liquid equilibria	VLE
vapor-liquid-liquid equilibria	VLLE
Pearson's chi-square test	χ^2
Extreme gradient boosting	XGBoost
Square of the regression correlation coefficient (absolute fraction of variance)	R_c^2

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